INTRODUCTION TO

EXPLAINABLE AI (XAI)

Making Artificial Intelligence
Transparent, Interpretable and Trustworthy



TRANSPARENCY
Understand AI Decisions

TRUST
Build Confidence in AI Systems

FAIRNESS
Ensure Ethical and Unbiased Outcomes

BLACK BOX PROBLEM
Complex models are accurate but hard to understand.

XAI SOLUTION
Explainable AI reveals the reasoning behind predictions.

BETTER DECISIONS
More transparency leads to better trust, safety and accountability.

HUMAN–AI COLLABORATION
XAI enables humans and AI to work together effectively.

FROM BLACK BOX TO EXPLAINABLE AI



XAI opens the black box and provides insights into how and why AI makes decisions.

Dr Kiran Jasthi

Unit I — Outline

01 Introduction to XAI

02 Goals of XAI

03 Brief History of Interpretability

04 Purpose of XAI

05 Societal Impact

06 Types of Explanations

07 Trade-offs

08 Taxonomy of XAI Techniques

09 Flow chart

01

Introduction to XAI

What is Explainable AI?

- Explainable AI (XAI) refers to methods and techniques that make the outputs and internal workings of AI/ML systems understandable to humans.
- As models grow in complexity (deep nets, ensembles), their decision logic becomes opaque to users, regulators, and even developers.
- XAI bridges the gap between high predictive accuracy and human comprehensibility.
- Two related but distinct terms: **Interpretability** (can a human understand the mechanism?) and **Explainability** (can the model's decision be justified post-hoc?).



THE BLACK BOX PROBLEM

THE BLACK BOX PROBLEM

AI models make accurate predictions, but the reasoning behind them is hidden and difficult for humans to understand.



WHY IS IT A PROBLEM?

LACK OF TRANSPARENCY
We don't know how or why a decision was made.

LACK OF TRUST
Users and stakeholders may not trust the model.

DIFFICULT DEBUGGING
Hard to identify errors or improve the model.

BIAS & UNFAIRNESS RISK
Hidden biases in data or model may lead to unfair outcomes.

REGULATORY CHALLENGES
Many regulations require explainable decisions.

HOW THE BLACK BOX WORKS



INPUT DATA
Raw data is fed into the model.

COMPLEX PROCESSING
Millions of parameters and layers process the information.

HIDDEN REASONING
Internal logic and patterns are not interpretable.

OUTPUT PREDICTION
The model gives a prediction or decision.

REAL-WORLD EXAMPLE

Loan Approval System



? But why was the loan rejected? Which factors influenced the decision?

WHERE IT MATTERS

- Healthcare (Diagnosis)
- Finance (Loan Approval, Fraud Detection)
- Autonomous Vehicles
- Recruitment & HR
- Criminal Justice
- And many more...

> "The goal of AI should not only be accuracy, but also **understandability**."



THE CONFUSION MATRIX

UNDERSTANDING MODEL PERFORMANCE

A confusion matrix evaluates how well a classification model performs by comparing the model's predictions with the actual class labels. It summarizes correct and incorrect predictions, allowing us to compute metrics such as accuracy, precision, recall, and F1-score.

INPUT

- Actual Labels (Ground Truth)
- Predicted Labels (Model Output)
- Classification Model

CONFUSION MATRIX

	PREDICTED POSITIVE	PREDICTED NEGATIVE
ACTUAL CLASS	TRUE POSITIVE (TP) ✓ Correctly predicted positive	FALSE NEGATIVE (FN) ✗ Missed positive (Type-II Error)
ACTUAL NEGATIVE	FALSE POSITIVE (FP) ✗ Wrongly predicted positive (Type-I Error)	TRUE NEGATIVE (TN) ✓ Correctly predicted negative

OUTPUT

- Accuracy
- Precision
- Recall
- F1-Score
- Model Evaluation

WHAT DOES EACH TERM MEAN?

TRUE POSITIVE (TP)
Model correctly predicts the positive class.
Example: Fraud transaction correctly detected as fraud.

TRUE NEGATIVE (TN)
Model correctly predicts the negative class.
Example: Legitimate transaction correctly classified as normal.

FALSE POSITIVE (FP)
Model predicts positive although it is actually negative.
Example: Genuine customer is wrongly flagged as fraud.
Also called: Type-I Error

FALSE NEGATIVE (FN)
Model predicts negative although it is actually positive.
Example: Fraud transaction is missed.
Also called: Type-II Error

HOW THE CONFUSION MATRIX WORKS



ACTUAL LABELS
Ground truth from the dataset

AI MODEL
Model makes predictions

PREDICTED LABELS
Model output for each sample

CONFUSION MATRIX
Compare predictions with actual labels

PERFORMANCE METRICS
Evaluate model performance

PERFORMANCE METRICS

ACCURACY

$$\frac{TP + TN}{TP + TN + FP + FN}$$

Overall percentage of correct predictions.

PRECISION

$$\frac{TP}{TP + FP}$$

How many predicted positives are actually positive.

RECALL (SENSITIVITY)

$$\frac{TP}{TP + FN}$$

How many actual positives are correctly identified.

F1 SCORE

$$\frac{2 \times Precision \times Recall}{Precision + Recall}$$

Balances Precision and Recall.

REAL-WORLD EXAMPLE

Medical Diagnosis

CONFUSION MATRIX

	Predicted Positive	Predicted Negative
Actual Positive	40	8
Actual Negative	5	147

RESULTS
- TP = 40
- FP = 5
- FN = 8
- TN = 147

METRICS

Accuracy	93.5%
Precision	88.9%
Recall	83.3%
F1 Score	86.0%

The model detects 40 actual disease cases correctly, misses 8 cases, and raises 5 false alarms.



A confusion matrix tells us **how** well a model performs, but it does not explain **why** a particular prediction was made.

This is where Explainable AI (XAI) complements the confusion matrix by revealing the reasoning behind individual predictions.

WHY IS IT IMPORTANT?

- Evaluates classifier performance
- Detects false alarms and missed detections
- Useful for imbalanced datasets
- Supports model comparison
- Widely used in Healthcare, Finance, Cybersecurity, and Power Systems

REAL-WORLD EXAMPLE

Medical Diagnosis – Heart Disease Detection

SCENARIO

An AI model is tested on 200 patients to detect heart disease.

Total Patients	200
Patients with Disease	48
Healthy Patients	152

CONFUSION MATRIX

ACTUAL \ PREDICTED	Disease (Predicted Positive)	Healthy (Predicted Negative)
Disease (Actual Positive)	**TRUE POSITIVE (TP)** 40 — Correctly detected disease	**FALSE NEGATIVE (FN)** 8 — Missed disease (Type-II Error)
Healthy (Actual Negative)	**FALSE POSITIVE (FP)** 5 — False alarm – healthy predicted as disease (Type-I Error)	**TRUE NEGATIVE (TN)** 147 — Correctly identified healthy

PERFORMANCE METRICS

ACCURACY	$\frac{TP + TN}{Total}$	$\frac{187}{200}$	93.5%
PRECISION	$\frac{TP}{TP + FP}$	$\frac{40}{40 + 5}$	88.9%
RECALL (SENSITIVITY)	$\frac{TP}{TP + FN}$	$\frac{40}{40 + 8}$	83.3%
F1 SCORE	$\frac{2 \times P \times R}{P + R}$	$\frac{2 \times 0.889 \times 0.833}{0.889 + 0.833}$	86.0%

INTERPRETATION

TRUE POSITIVE (40)
40 patients have heart disease and the model correctly identified them.

FALSE NEGATIVE (8)
8 patients have heart disease, but the model missed them. (Dangerous Error)

FALSE POSITIVE (5)
5 healthy patients were incorrectly predicted as having heart disease. (False Alarm)

TRUE NEGATIVE (147)
147 healthy patients were correctly predicted as healthy.

RESULT SUMMARY



- Correct Predictions (TP + TN) **187** (93.5%)
- Errors (FP + FN) **13** (6.5%)
- Total 200 (100%)



KEY INSIGHT

The confusion matrix tells us how well the model performs. It does not explain why the model made mistakes.

Explainable AI (XAI) helps us understand why 8 patients were missed and why 5 healthy patients were flagged incorrectly, by revealing the most important features (e.g., age, cholesterol, ECG, blood pressure, etc.) that influenced those predictions.

WHY THIS MATTERS?

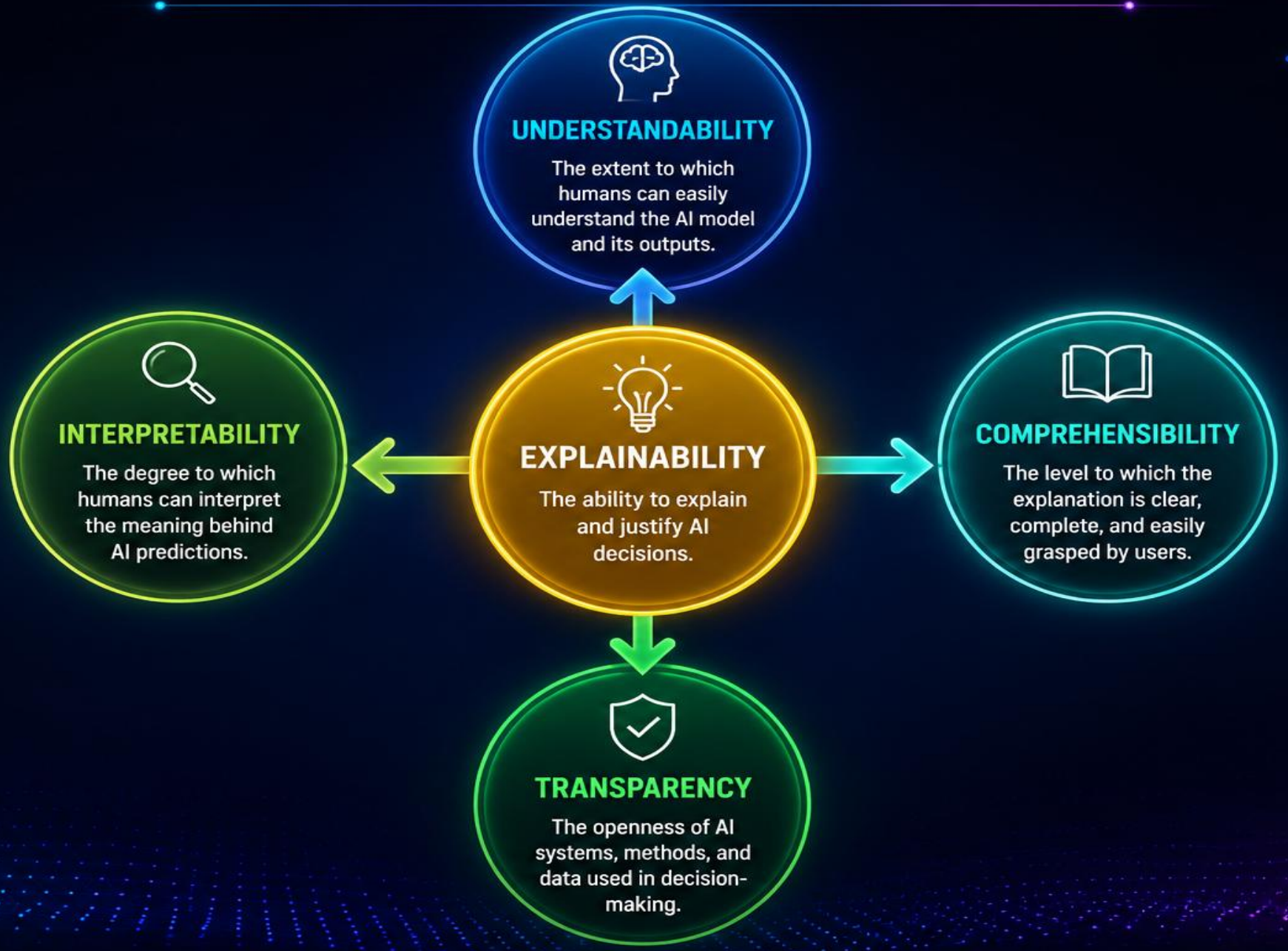
- Improves trust in AI decisions
- Helps doctors make confident clinical decisions
- Reduces missed diagnosis and false alarms
- Supports better healthcare outcomes

Confusion Matrix evaluates performance. Explainable AI provides the reason behind each prediction

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Goals of XAI

THE CORE PRINCIPLES OF EXPLAINABLE AI (XAI)



Stakeholders Driving These Goals

- **Data Scientists / Developers:** Need explanations to debug, validate, and improve models.
- **Domain Experts:** Need explanations to verify decisions align with domain knowledge.
- **End Users:** Need simple, actionable explanations to trust and act on predictions.
- **Regulators / Auditors:** Need explanations to ensure compliance and fairness.
- **Business / Management:** Need explanations to justify decisions and manage risk.



03

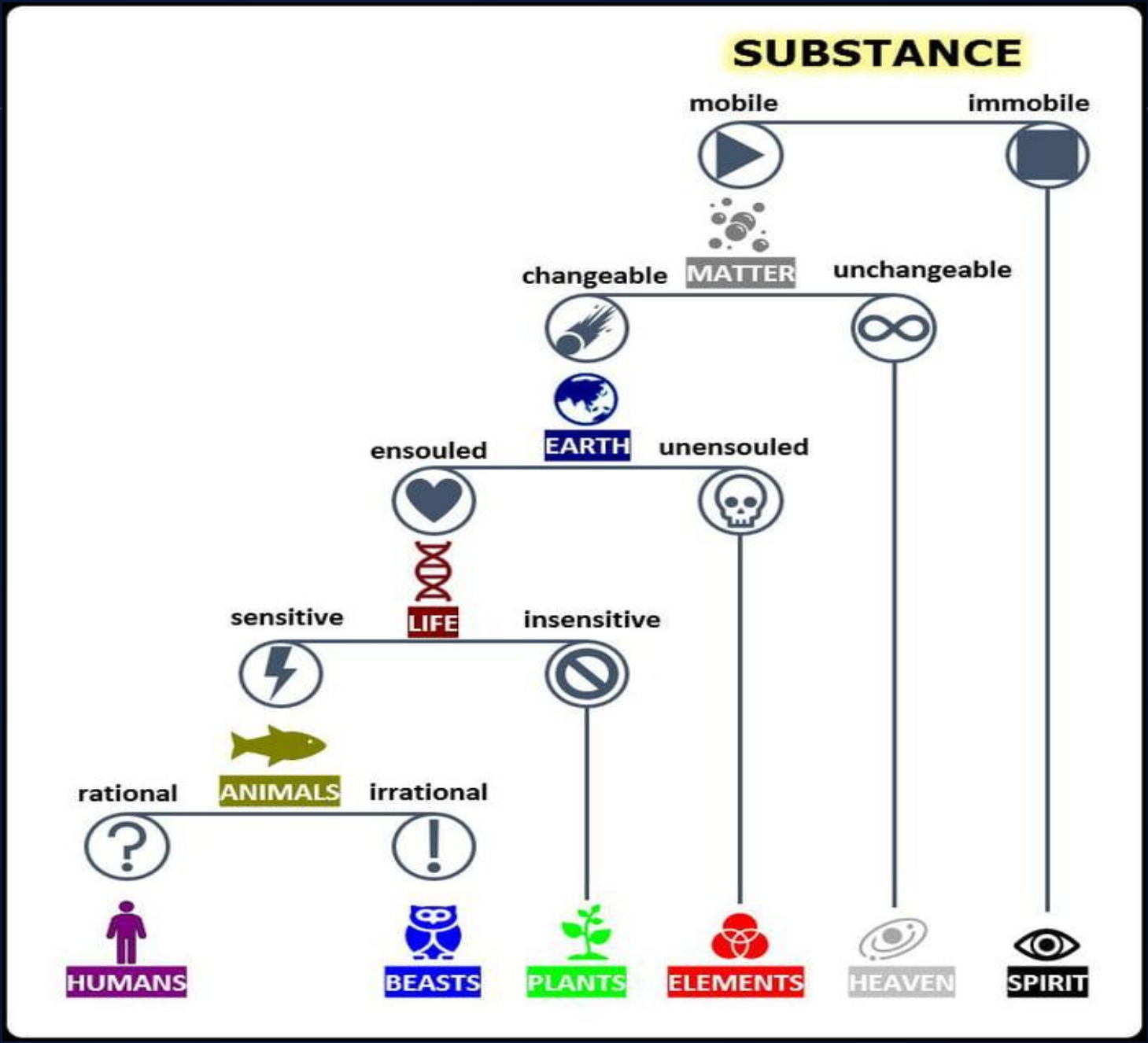
Brief History of Interpretability

A Brief History of Interpretability

- Interpretability is not a new concern -- it predates modern deep learning by centuries.
- Historical milestones include:
  - Porphyrian Trees (3rd century logic)
  - Expert Systems (1970s--80s)
  - Case-Based Reasoning (1980s)
  - Bayesian Networks (1980s--90s)
  - Neural Networks and the resurgence of the black-box debate (2010s)

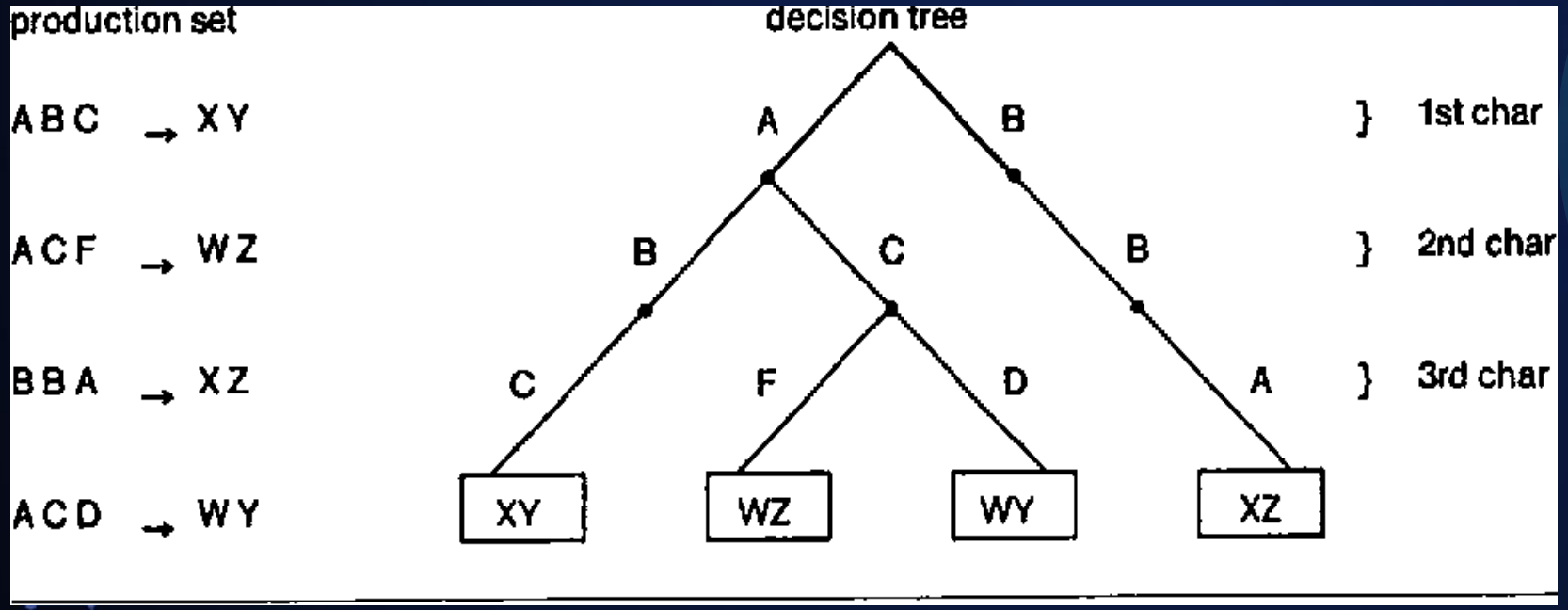


Porphyrian Tree



Expert Systems

- Rule-based AI systems (1970s--80s) such as MYCIN and DENDRAL encoded human expert knowledge as IF-THEN rules.
- Inherently interpretable: a chain of fired rules constitutes the explanation.
- Limitations: rule explosion, brittleness, difficulty in knowledge acquisition (knowledge engineering bottleneck).
- Established the precedent that an AI system should be able to justify its conclusions to the us



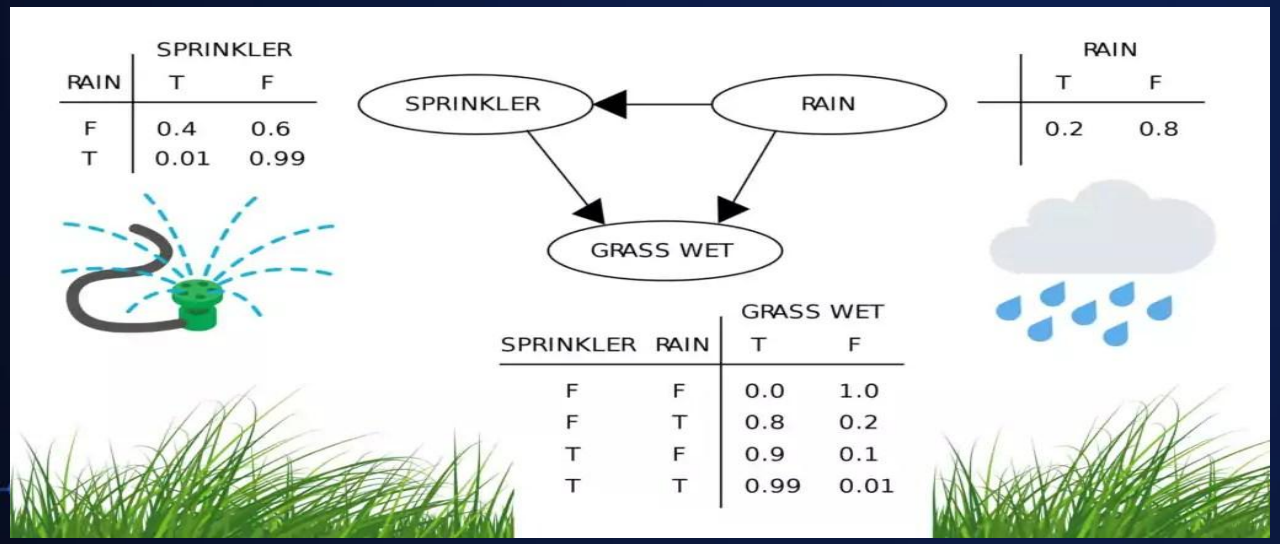
Case-Based Reasoning (CBR)

- Solves new problems by retrieving and adapting solutions to similar past cases.
- Explanation is natural: "This decision was made because it resembles past case X."
- Cycle: Retrieve → Reuse → Revise → Retain.
- Forms the conceptual basis for modern example-based explanation methods (e.g., k-NN explanations, prototypes).



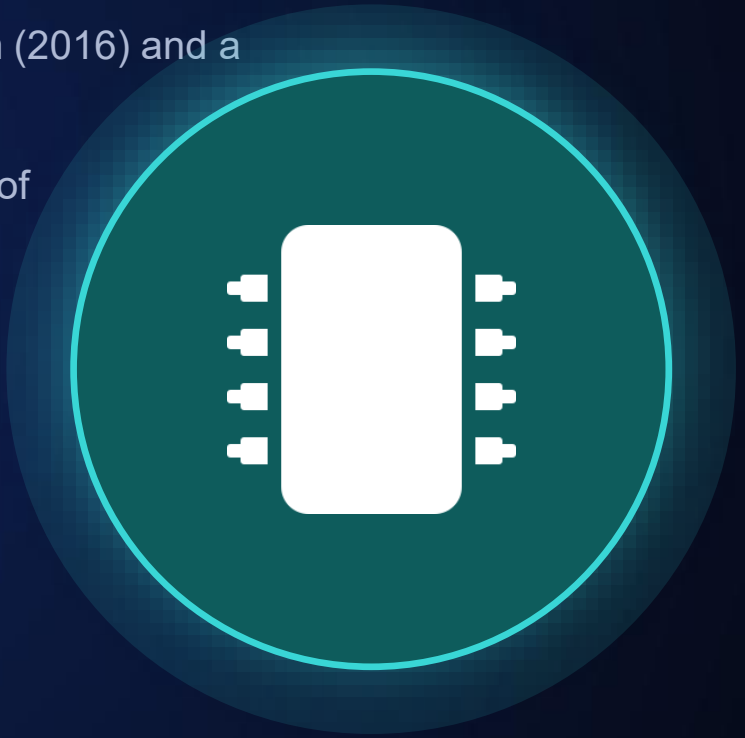
Bayesian Networks

- Probabilistic graphical models representing variables and their conditional dependencies via a directed acyclic graph (DAG).
- Each node's probability depends only on its parents -- enabling transparent probabilistic reasoning.
- Naturally support causal-style explanations: "Evidence at node A increased the probability of outcome B because of dependency A → B."
- Still widely used in medical diagnosis and risk assessment due to their interpretability.



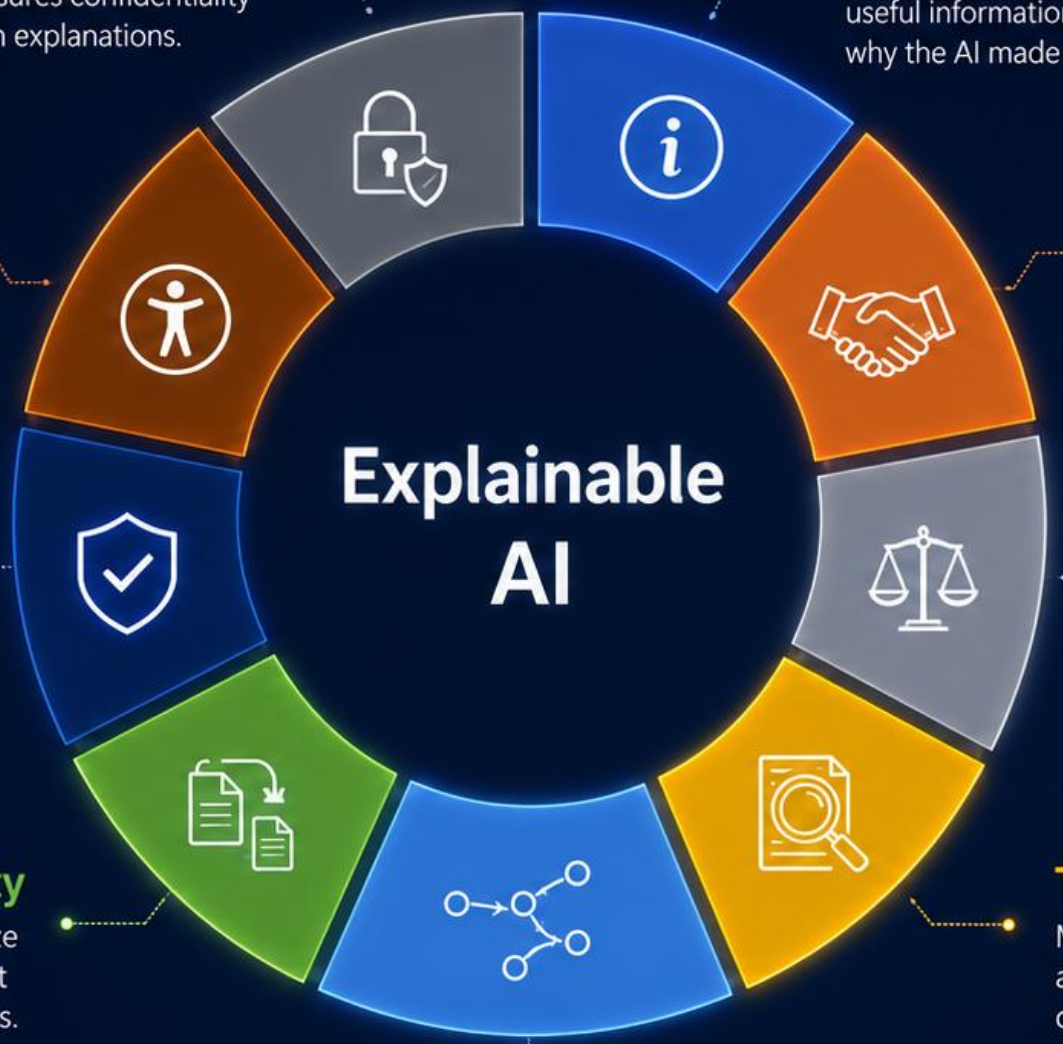
Neural Networks and the Modern XAI Era

- Deep neural networks (2010s) achieved breakthrough accuracy in vision, speech, and language tasks.
- However, millions of weighted connections make manual interpretation infeasible.
- This re-ignited interest in interpretability research, leading to DARPA's XAI program (2016) and a surge of post-hoc explanation techniques (LIME, SHAP, Grad-CAM, etc.).
- Modern XAI seeks to retain deep learning's accuracy while restoring some degree of transparency.



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Purpose of XAI



Privacy

Protects sensitive data and ensures confidentiality in explanations.

Informativeness

Provides relevant and useful information about why the AI made a decision.

Accessibility

Ensures explanations are easy to understand and usable by the target users.

Trustworthiness

Builds user trust through dependable, truthful and credible explanations.

Reliability

Produces consistent and stable explanations under similar conditions.

Fairness

Ensures unbiased and impartial decisions for all individuals.

Transferability

Explanations generalize well across different contexts and domains.

Transparency

Makes the model, data and decision process clear and open.

Causality

Reveals cause-and-effect relationships, not just correlations.

INFORMATIVENESS

Useful, relevant information explaining why the AI made a decision.

EXAMPLE

Loan AI highlights income and credit score as the main factors.

TRUSTWORTHINESS

Builds confidence in the AI system and its explanations.

EXAMPLE

A doctor reviews the evidence supporting an AI diagnosis.

TRANSPARENCY

Shows how decisions are made and which factors influence the output.

EXAMPLE

A prediction model displays the most influential features.

PURPOSE OF EXPLAINABLE AI

FAIRNESS

Avoids unjustified bias or discrimination in AI decisions.

EXAMPLE

A hiring model evaluates candidates without unfair gender or ethnicity bias.

RELIABILITY

Produces consistent and dependable outputs under similar conditions.

EXAMPLE

Similar inputs produce similar predictions and explanations.

PRIVACY

Explains decisions while protecting sensitive or confidential information.

EXAMPLE

Medical AI explains a result without exposing another patient's data.

PURPOSE OF EXPLAINABLE AI

CAUSALITY

Distinguishes genuine cause-and-effect from simple correlation.

EXAMPLE

Higher temperature causes increased cooling demand.

TRANSFERABILITY

Keeps explanations meaningful across related situations or domains.

EXAMPLE

An XAI method works across power-grid datasets from different regions.

ACCESSIBILITY

Presents explanations in a form suitable for the intended user.

EXAMPLE

Engineers see feature importance; consumers receive plain-language reasoning.

PURPOSE OF EXPLAINABLE AI

THE TRUST FORMULA IN XAI

Trust in AI Systems is Built on Four Key Pillars

$$T = \frac{C + R + I}{S}$$

T — Trust

C — Credibility

R — Reliability

I — Intimacy

S — Self-orientation

T – Trust

The confidence users have in an AI system and its decisions.

C – Credibility

The belief that the AI system is truthful, expert, and provides credible explanations.

R – Reliability

The expectation that the AI system performs consistently and dependably over time.

I – Intimacy

The sense of connection and understanding between users and the AI system.

S – Self-orientation

The extent to which users focus on their own goals, needs, and perspectives.

Interpretation: Trust increases when Credibility, Reliability, and Intimacy are high, and users are not excessively **self-oriented**.

05

Societal Impact

SOCIETAL IMPACT OF EXPLAINABLE AI

Building trustworthy AI for a fairer, safer, and more accountable society

1 FAIRNESS & BIAS

AI can inherit and amplify biases, leading to unfair and discriminatory outcomes.



How XAI Helps

Uncovers hidden biases in data and models, enabling detection, mitigation and fairer decisions.

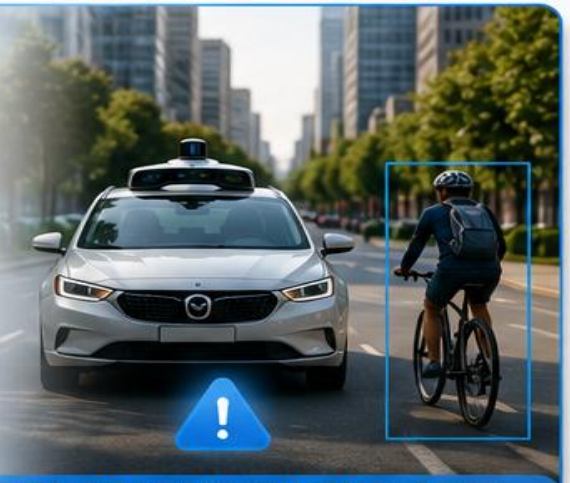
Real-World Example

COMPAS recidivism algorithm showed racial and gender bias in predicting re-offending risk.

Supports equality and promotes justice in AI-driven decisions.

2 SAFETY

AI systems must not make decisions or take actions that could harm humans, either accidentally or maliciously.



How XAI Helps

Identifies potential failure modes, blind spots and unsafe behaviors before they cause real-world harm.

Real-World Example

Autonomous vehicles use XAI to detect why a pedestrian or cyclist might not be recognized in certain conditions.

Protects human life and builds confidence in AI adoption.

EXPLAINABLE AI (XAI)

Turning Black-Box AI into Trustworthy Intelligence

3 VERIFIABILITY

Testing alone cannot guarantee that an AI system behaves as intended in every possible situation.



How XAI Helps

Makes model behavior interpretable so it can be verified, bounded and validated using formal methods.

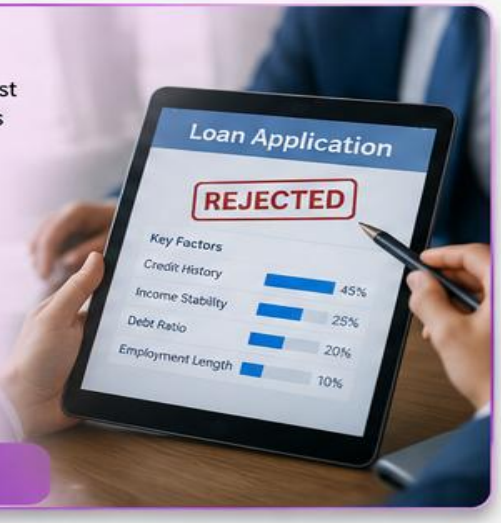
Real-World Example

In medical AI, verification ensures that treatment recommendations stay within safe and reliable boundaries.

Ensures reliability and robustness in critical decision-making.

4 ACCOUNTABILITY

When AI makes a decision, we must know why it happened and who is responsible.



How XAI Helps

Provides clear explanations for individual predictions and overall model behavior, supporting transparency and responsibility.

Real-World Example

A bank can explain why a loan was rejected and demonstrate that the decision was based on relevant and fair factors.

Enables responsibility, traceability and ethical governance.

THE BIG PICTURE

XAI helps create AI systems that are **FAIR**, **SAFE**, **VERIFIABLE** and **ACCOUNTABLE** — driving **trust**, **empowering** people and delivering **positive impact** for society.

Fair | Safe | Verifiable | Accountable

06

Types of Explanations

Types of Explanations

- Global explanations
- Local explanations
- Contrastive Explanations
- What if explanations
- Counterfactual explanations
- Example-based explanations



TYPES OF EXPLANATIONS

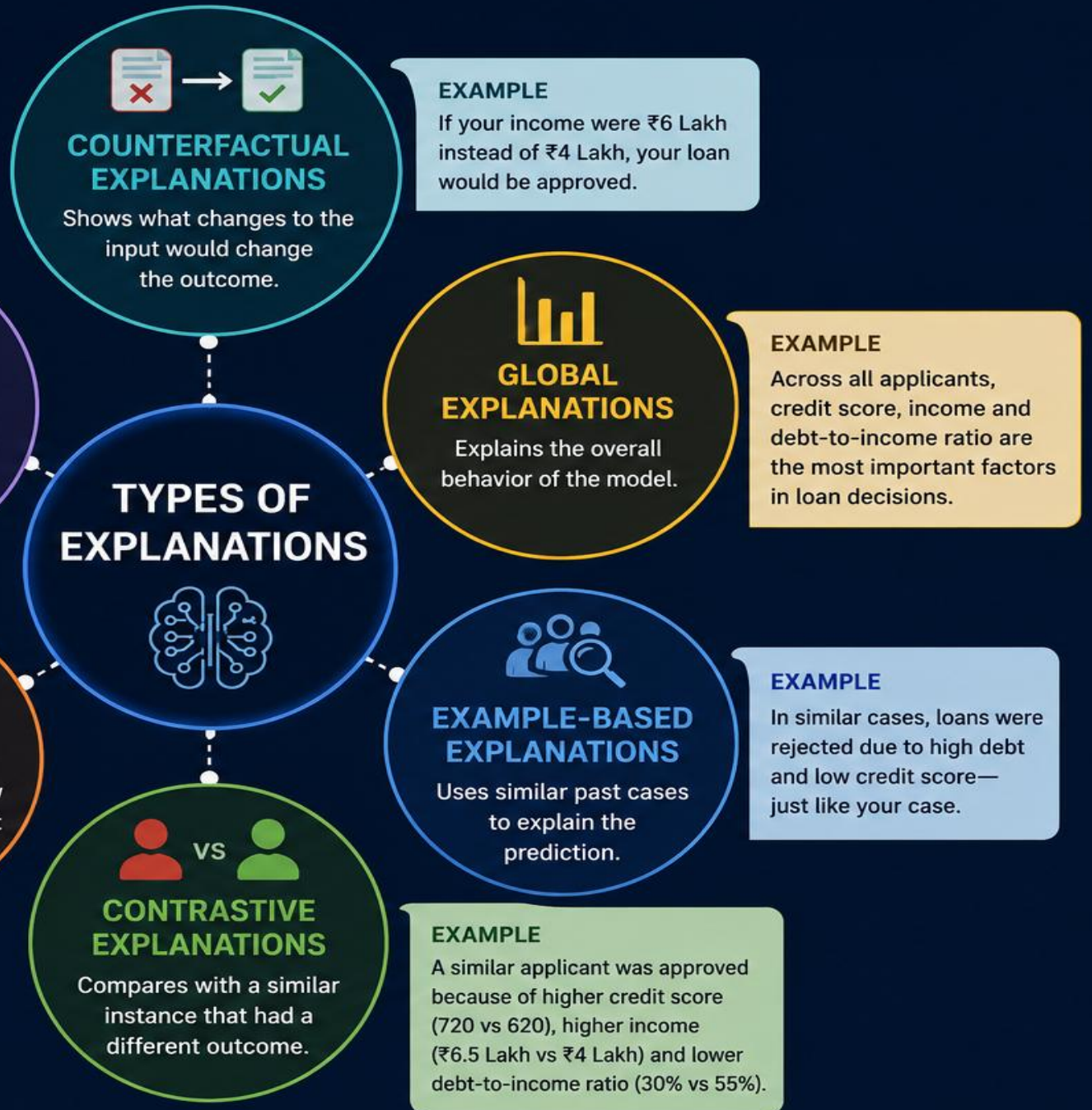
Different ways XAI explains AI decisions with a real-life example

SCENARIO

AI model is used by a bank to decide whether to approve a loan application.

BENEFITS

- ✓ Builds Trust
- ✓ Improves Transparency
- ✓ Reduces Bias
- ✓ Supports Better Decision-Making
- ✓ Ensures Accountability

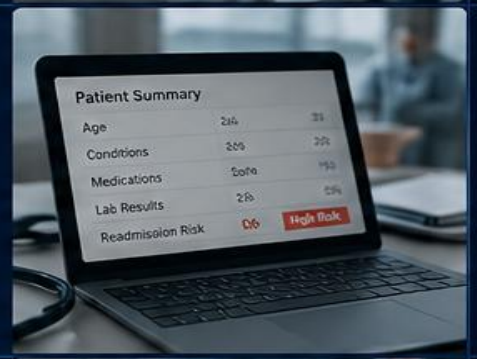


07

Trade-offs

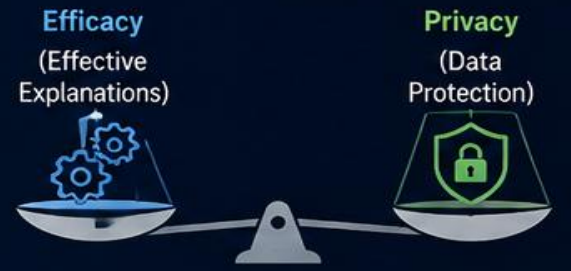
Key Trade-offs in Explainable AI (XAI)

In real-world AI systems, improving one aspect often comes at the cost of another.

Trade-off	What is the Trade-off?	Real-world Example	Visualization
1 Completeness vs Interpretability	• Complex models (e.g., deep neural networks) provide high accuracy and complete coverage but are hard to interpret. • Simple models (e.g., decision trees, linear models) are easy to interpret but may miss important patterns.	**Medical Diagnosis** A deep learning model detects cancer from images with 95% accuracy but is a "black box". A decision tree is easy for doctors to understand but achieves only 75% accuracy.	
2 Efficacy vs Privacy	• Detailed explanations can unintentionally leak sensitive information from training data. • Protecting privacy (e.g., via anonymization, encryption, or privacy-preserving ML) can reduce explanation detail and effectiveness.	**Healthcare (Patient Data)** An AI model predicts patient readmission risk. A detailed explanation may reveal sensitive disease or treatment information. Applying privacy filters protects patient data but may remove important context.	
3 Human Understandability vs Accuracy	• Highly accurate models often capture complex, non-linear relationships that are difficult for humans to understand. • Simplified explanations (e.g., linear approximations or surrogate models) are easier to understand but may not fully represent the model.	**Loan Approval** A model using hundreds of features achieves 92% accuracy in predicting loan default. A simple rule-based explanation (e.g., "high debt-to-income ratio") is easy to understand but doesn't capture all the factors the model uses.	



Hard to achieve both at the same time



More privacy often means less explanatory detail



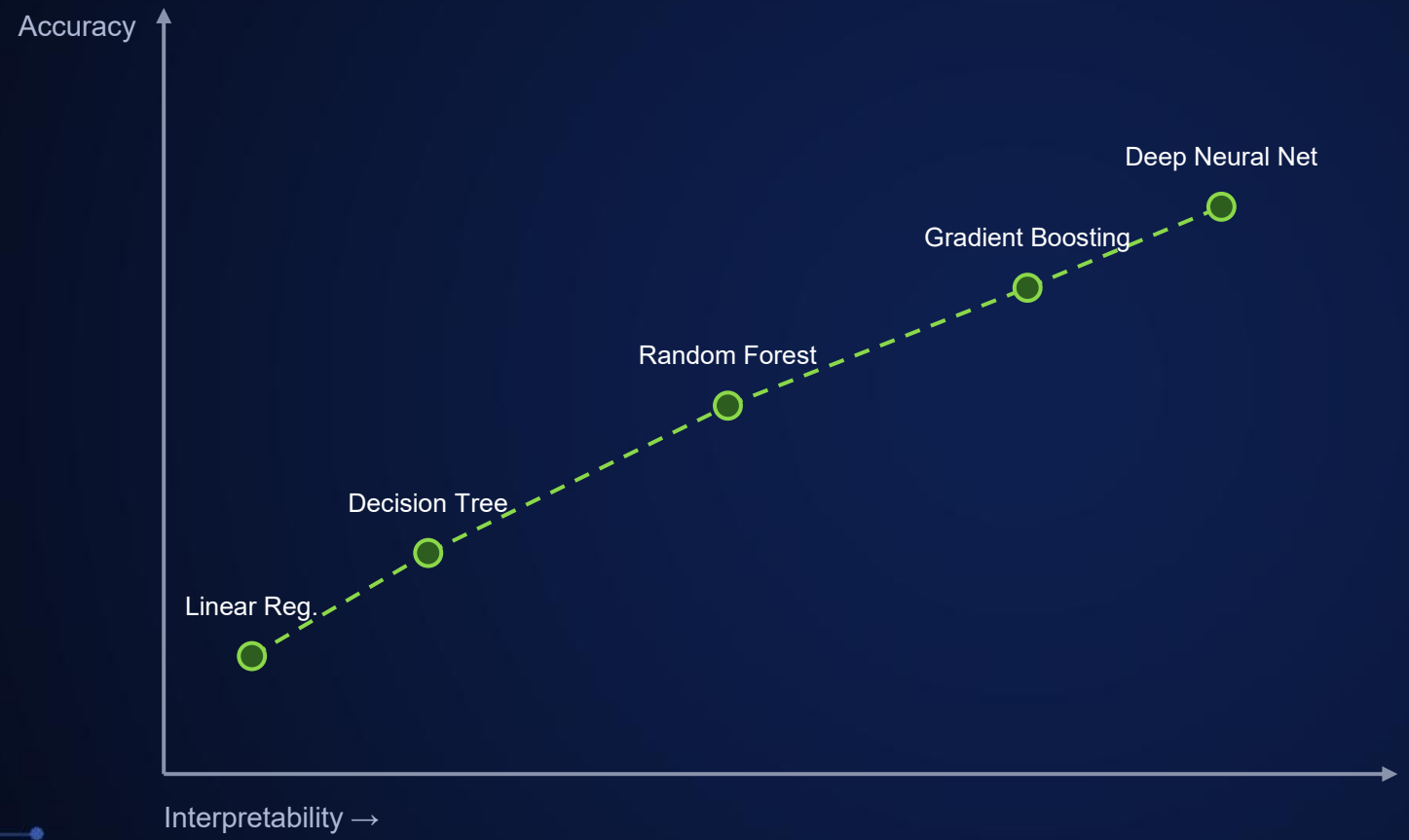
Simpler explanations may reduce accuracy

The Goal of XAI: Find the right balance between these trade-offs to build AI systems that are accurate, trustworthy, interpretable, and privacy-preserving.

Why it Matters

- Trust & Adoption
- Regulatory Compliance
- Fairness
- Better Decision-Making

Visualizing the Accuracy-Interpretability Trade-off



- Note: this trade-off is not absolute -- some interpretable models (e.g., EBMs) can rival black-box accuracy.

08

Taxonomy of XAI Techniques

Taxonomy of XAI Techniques

- XAI methods can be organized along two key axes:
  - **Scope** -- Global vs. Local
  - **Stage** -- Pre-model, In-model (intrinsic), Post-model (post-hoc)
  - Model -- model-specific vs. model-agnostic



Explainable AI

Making AI decisions understandable, transparent and trustworthy

Scope

The level at which explanations are generated

Global

Explains the behavior of the model as a whole



Example

Feature importance showing which factors influence the model's predictions overall.

Local

Explains the prediction for a specific instance

Loan Application #4521	REJECTED
High Debt-to-Income	+0.42
Low Credit Score	+0.31
Short Credit History	+0.17
Stable Income	−0.21
Employed	−0.12

Example

Why this particular loan application was rejected.

Stage

The stage of the AI pipeline where explanations are applied

Pre

Before the model is built

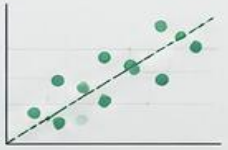


Example

Data preprocessing, feature selection, data visualization

Intrinsic

During model building



Example

Interpretable models like Linear Regression, Decision Trees

Post-Hoc

After the model makes a prediction



Example

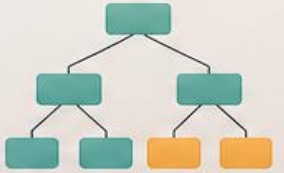
SHAP values, LIME, Counterfactual Explanations

Model

The relationship between the explanation method and the model

Specific

Designed for specific types of models

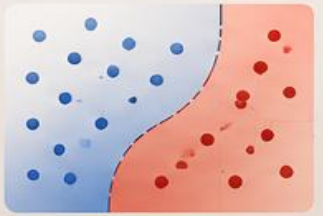


Example

Decision Trees, Rule Lists (Explains models that are interpretable by design)

Agnostic

Model-agnostic methods (works on any model)



Example

SHAP, LIME, Anchors (Can explain any black-box model)

Why This Taxonomy Matters?

Helps practitioners choose the right explanation method based on the problem, model, and purpose.

- Improves Transparency & Trust
- Supports Regulatory Compliance
- Enhances Human-AI Collaboration
- Enables Better Decision-Making

09

Flowchart for Interpretable and Explainable Techniques

EXPLAINABLE AI (XAI)

Making AI Decisions Understandable, Transparent & Trustworthy

EXPLAIN DATA
Why the data matters

EDA METHOD
Understand & visualize the data

GRAPHICAL
- Multivariate
- Pair Plot
- Heatmap
- PDP Plot
- Joint Plot
- Radar Chart
- Projection Plot
- Parallel Coord Plot

NON-GRAPHICAL
- Univariate
- Box Plot
- Density Plot
- Violin Plot

STAT ANALYSIS
Statistical Analysis

FOCUS
Where to focus the explanation

MODEL TYPE
What type of model are we explaining?

NEURAL NETWORK
- Attention
- Jointly Trained

TRADITIONAL
- Linear Regression
- LR
- GLM
- Naïve Bayes

RULE-BASED
- Bayesian Logic
- Bayesian Case
- CORELS
- Bayesian Rule

INTRINSICALLY
Models that are interpretable by design

ENSEMBLE
- Boosted Rule-sets
- EBM
- RuleFit
- Stage-Rules

DECISION TREE
- Optimal CT
- Optimal DT
- DL8.5
- GOSDT

SCORING
- Linear Trigger
- Models

EXPLAIN MODEL
Why the model makes a decision

Treats any model as a black-box

INTERPRETATION METHOD
How do we generate the explanation?

These also work for neural networks

MODEL-AGNOSTIC
Model independent (Works for any model)

NEURAL NETWORK
- Perturbation
- LIME
- Occlusion
- RISE
- PNA
- Meaningful Perturbations

GENERAL
- Gradient
- Action Max
- Class Model Via Scoring Ctr
- DeepLIFT
- DeepSHAP
- Deconvolution
- Guided Backprop
- Integrated Grad
- LRP
- Excite Backprop
- CAM
- GradCAM
- TCAV

MODEL TYPE
Model specific (Depends on model)

VISUAL EXPLAIN
- PDP
- ICE
- cP
- ALE
- Breakdown
- iBreakdown

EXAMPLE-BASED
- Contrastive Exp
- kNN
- Trust Scores
- Counterfactuals
- Prototypes & Criticisms
- Influential Instances

FEATURE-BASED
- Interaction
- Permutation
- Ablation LOCO
- Shapley Values
- SHAP
- KernelSHAP
- Anchors
- Global Surrogate
- LIME

AT A GLANCE

Explain Data
Understand the data quality & patterns

Focus
Focus on the right part of the model or pipeline

Explain Model
Use the right method for clear explanations

Works for All Models
Model-agnostic methods work as black-box tools

Neural Network Ready
Specialized techniques for deep learning

BENEFITS

- Transparency & Trust
- Regulatory Compliance
- Better Decision Making
- Improved Model Reliability

10

Pre-model Interpretability and Explainability

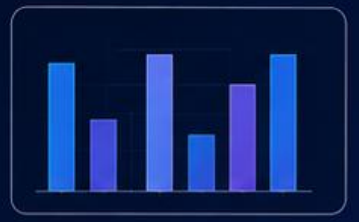
Pre-model Interpretability:

- Pre-model interpretability focuses on understanding and preparing data *before* any model is trained.
- Rationale: a model can only be as interpretable (and as good) as the data and features it is built upon.
- Two major components:
  - Data Science Process and Exploratory Data Analysis (EDA)
  - Feature Engineering

The Data Science Process

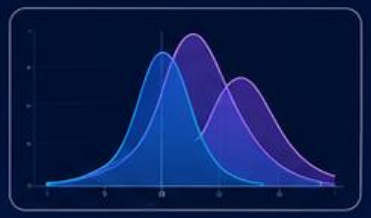
- Typical pipeline: **Problem Definition** → **Data Collection** → **EDA** → **Feature Engineering** → **Modeling** → **Evaluation** → **Deployment**.
- Explainability considerations should be embedded from the very first stage, not bolted on afterward.
- Decisions made early (data sources, sampling, labeling) strongly influence what can later be explained.

Data



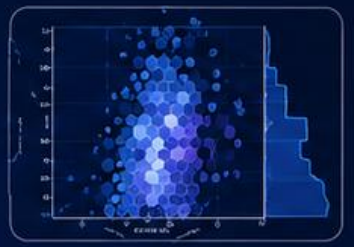
1. Distinguish Features

Identify and separate relevant variables



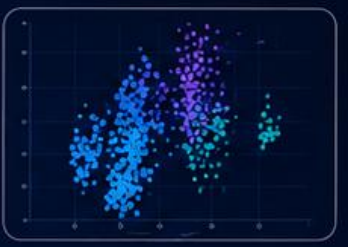
2. Univariate Analysis

Examine individual feature distributions



3. Bivariate and Multivariate Analysis

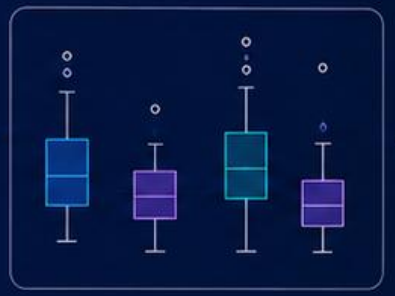
Explore relationships between features



4. Detect Aberrant and Missing Values

Identify anomalies and handle missing data

ID	Feature 1	Feature 2	...	Feature n
1	2.45	5.61	...	3.12
2	1.23	2.34	...	NaN
3	-0.56	1.02	...	4.77
...	...	...	...	...
n	0.85	-1.23	...	2.91



5. Detect Outlier

Spot outliers using statistical methods and visualizations



6. Feature Engineering

Transform, create and select features to improve models



Insights

- Better Decisions & Action
- Deeper Insights & Pattern Discovery
- Improved Performance & Business Outcomes

Exploratory Data Analysis (EDA)

- EDA is the process of summarizing, visualizing, and understanding a dataset's main characteristics before modeling.
- Goals: detect patterns, spot anomalies, test assumptions, and check data quality.
- Forms the foundation for explainability: understanding data distributions helps later explain model behavior in terms of those same distributions.

EDA Challenges in Explainable AI (XAI)

Data Challenges That Influence Model Interpretability

1. High Dimensionality



- Large number of features increases computational complexity and makes visualization difficult.

Impact:
Harder to identify influential features and generate interpretable explanations.

2. Mixed Data Types

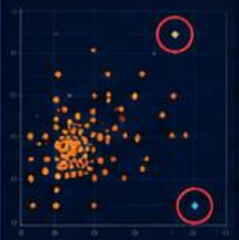
ID	Age	Gender	Income	Rating
101	25	Male	45,000	4
102	34	Female	67,500	5
103	28	Male	52,300	3
...	...	...	...	...



- Datasets contain numerical, categorical, ordinal, text, and temporal variables.

Impact:
Different preprocessing methods complicate interpretation and feature attribution.

3. Missing Values & Outliers



A	B	C	D
1.2	NaN	3.4	4.5
2.1	1.1	NaN	3.3
NaN	0.5	1.8	2.9
1.0	2.2	3.1	NaN

- Real-world data often contain incomplete records and abnormal observations.

Impact:
Can bias feature importance, SHAP values, and local explanations.

4. Privacy & Anonymization



- Sensitive information must be protected using anonymization or masking.

Impact:
Limited feature access may reduce explanation fidelity and transparency.

5. Massive Data Volume



- Modern datasets may contain millions or billions of records.

Impact:
Traditional EDA becomes computationally expensive, requiring sampling or aggregation that may overlook important patterns.

Why These Challenges Matter

Complex Data → Difficult Exploratory Data Analysis → Poor Feature Understanding → Less Reliable AI Model → Misleading Explanations → Reduced Trust in AI

Key Takeaway: Explainable AI begins with high-quality exploratory data analysis. Addressing dimensionality, heterogeneous data, missing values, privacy, and scalability ensures that explanations produced by XAI methods are accurate, reliable, and trustworthy.

EDA: Taxonomy

- EDA techniques can be categorized as:
  - **Non-graphical:** summary statistics (mean, median, variance, skewness).
  - **Graphical:** histograms, box plots, scatter plots, pair plots.
  - **Univariate:** examines one variable at a time.
  - **Multivariate:** examines relationships across multiple variables.

Exploratory Data Analysis (EDA) Techniques

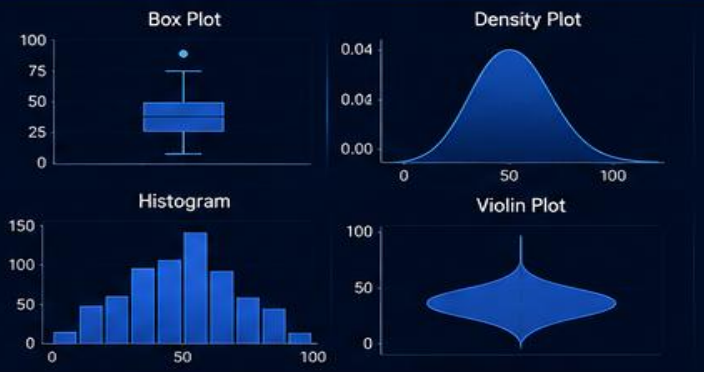
Classification of EDA Methods

GRAPHICAL METHODS

NON-GRAPHICAL METHODS

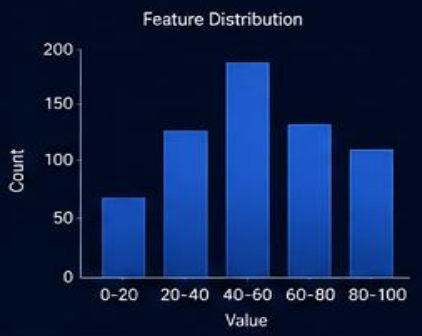
UNIVARIATE ANALYSIS

- Box Plots
- Density Plots
- Violin Plots
- Histograms



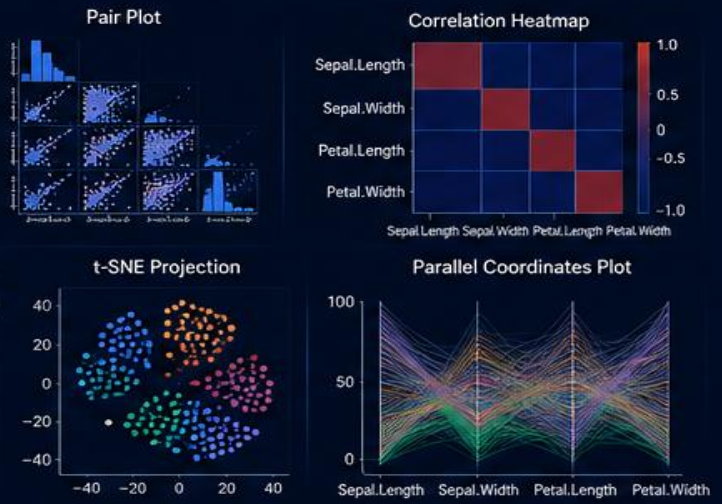
- Summary Statistics

Statistic	Value
Count	1,000
Mean	54.23
Median	53.06
Standard Deviation	16.76
Variance	281.04
Skewness	0.23
Kurtosis	-0.41
Minimum	5.00
Maximum	98.00
Missing Values	0 (0.00%)



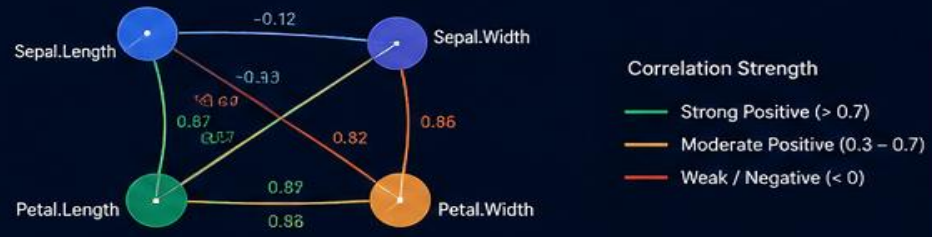
MULTIVARIATE ANALYSIS

- Pair Plots
- Heatmaps
- Biplots
- Joint Plots
- Radial Charts
- Projection Plots (T-SNE, MDS, etc.)
- Parallel Coordinates



- Correlation Analysis

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width
Sepal.Length	1.00	-0.12	0.87	0.82
Sepal.Width	-0.12	1.00	-0.43	-0.37
Petal.Length	0.87	-0.43	1.00	0.96
Petal.Width	0.82	-0.37	0.96	1.00



Raw Data → EDA Techniques → Feature Understanding → Explainable AI (XAI) → Trustworthy Decisions

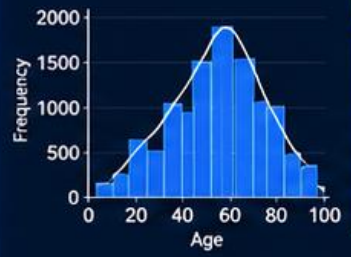
Role of EDA in Explainability

- Reveals data quality issues (missing values, duplicates, inconsistent encodings) before they propagate into the model.
- Helps identify which features are likely to be predictive or redundant.
- Establishes baseline distributions used later to judge whether model explanations (e.g., feature importance) make sense.
- Encourages a transparent, well-documented data understanding process -- itself a form of pre-model interpretability.

Role of EDA in Explainability & Interpretability

EDA provides the foundation for trustworthy, interpretable and reliable AI models.

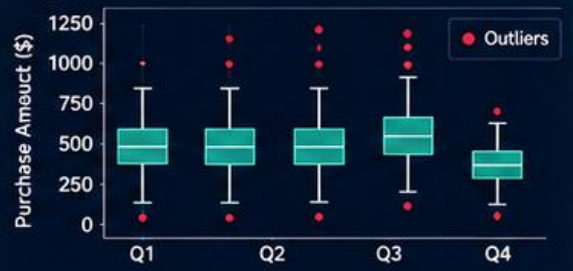
01 Understand Data Characteristics



Summary Statistics	
Count	10,000
Mean	54.23
Median	53.06
Std Dev	16.76
Min	5.00
Max	98.00
Missing	2.1%

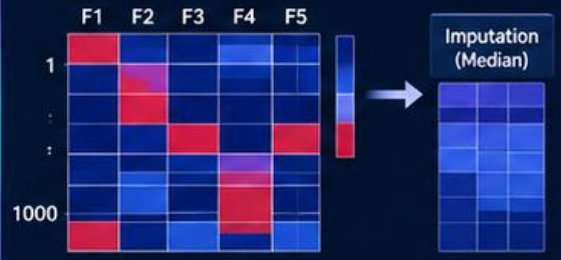
- Analyze spread, ranges, distributions
- Identify missing values using univariate statistics

02 Detect & Handle Outliers



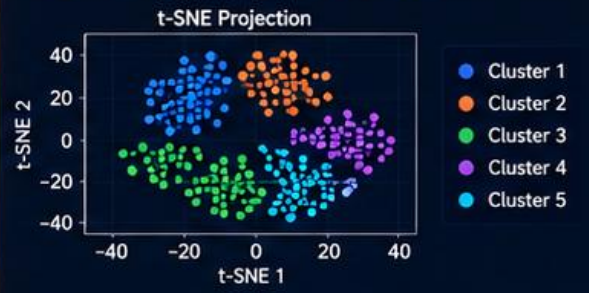
- Detect outliers using distribution & box/violin plots
- Overcome issues of outliers in modeling

03 Handle Missing Values



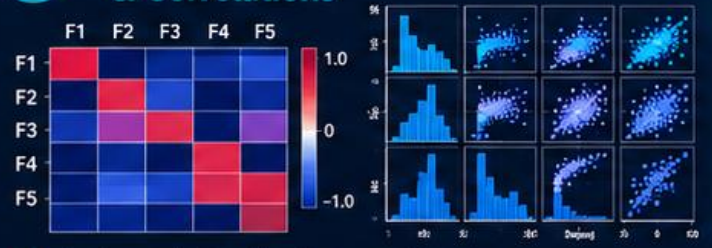
- Discover missing values
- Impute strategically to gain insights and improve data quality

04 Project to Lower Dimensions & Visualize Patterns



- Project data to lower dimensions (PCA, t-SNE, MDS, etc.)
- Discover local and global patterns

05 Discover Relationships & Correlations



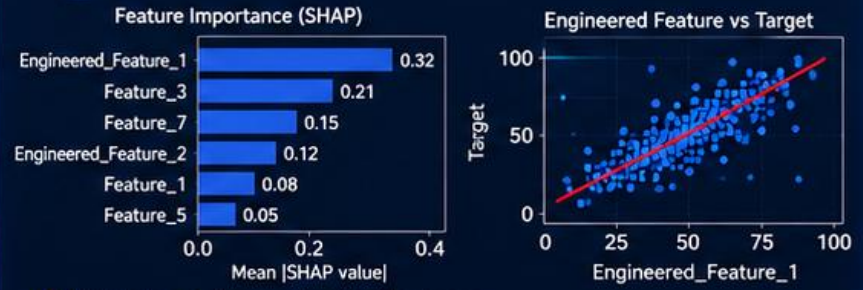
- Analyze relationships between features
- Correlation analysis supports global and local pattern understanding

06 Validate Expert Hypotheses



- Validate domain expert hypotheses
- Confirm patterns and scenario understanding

07 Validate New / Engineered Features



- Validate newly constructed features
- Check local & global interactions and their impact

Raw Data → Exploratory Data Analysis → Better Feature Understanding → Explainable AI (XAI) → Trustworthy Decisions

Non-graphical EDA: Summary Statistics

- **Measures of central tendency:** mean, median, mode.
- **Measures of spread:** variance, standard deviation, interquartile range (IQR).
- **Shape statistics:** skewness (asymmetry) and kurtosis (tailedness).
- **Categorical summaries:** frequency counts, mode, contingency tables.
- These statistics provide a quick, interpretable numerical snapshot of each variable.

DESCRIPTIVE STATISTICS IN EDA

A quick, interpretable numerical snapshot of each variable

1. MEASURES OF CENTRAL TENDENCY

- Mean: average of all values
- Median: middle value when sorted
- Mode: most frequently occurring value

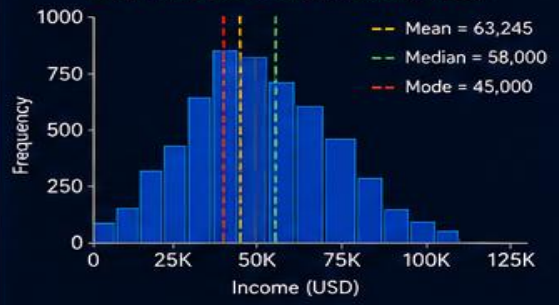
EQUATIONS

Mean ($\bar{x}$): $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$

Median: middle value of ordered data

Mode: value with highest frequency

EXAMPLE: Distribution of Income (USD)



Summary Statistics (Income)				
Count	Mean	Median	Mode	Std Dev
10,000	63,245	58,000	45,000	16,760

2. MEASURES OF SPREAD

- Variance: average squared deviation from mean
- Standard Deviation: square root of variance
- Interquartile Range (IQR): Q3 – Q1

EQUATIONS

Variance (σ^2): $\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$

Std. Deviation (σ): $\sigma = \sqrt{\sigma^2}$

IQR = $Q_3 - Q_1$

EXAMPLE: Box Plot of Income (USD)



Summary Statistics (Income)				
Variance	Std Dev	Q1 (25%)	Q3 (75%)	IQR
281,044,900	16,760	40,000	75,000	35,000

3. SHAPE STATISTICS

- Skewness: degree of asymmetry
- Kurtosis: degree of tailedness (peakedness)

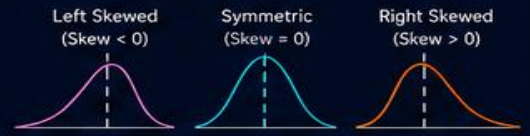
EQUATIONS

Skewness (g_1): $g_1 = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{s}\right)^3$

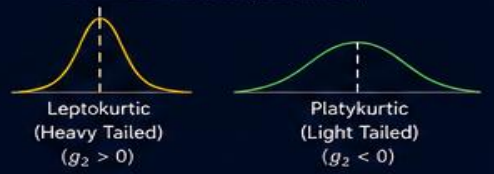
Kurtosis (g_2): $g_2 = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{s}\right)^4 - 3$

where s = standard deviation

EXAMPLE: Distribution Shapes (Skewness)



EXAMPLE: Kurtosis (Tailedness)



Shape Statistics (Income)	
Skewness (g_1)	Kurtosis (g_2)
0.62	0.48

4. CATEGORICAL SUMMARIES

- Frequency Counts: number of occurrences
- Mode: most frequent category
- Contingency Tables: relationship between categorical variables

EXAMPLE: Frequency Table (Education Level)

Education Level	Count	Percentage
High School	1,250	25.4%
Bachelor's	2,180	44.3%
Master's	1,230	25.0%
PhD	260	5.3%
Total	4,920	100.0%

EXAMPLE: Frequency Chart



EXAMPLE: Contingency Table (Gender vs Purchase)

	Gender		
Purchase	Male	Female	Total
Yes	850	950	1,800
No	650	470	1,120
Total	1,500	1,420	2,920

SUMMARY OF KEY EQUATIONS

Mean	Median	Mode	Variance	Std. Deviation	IQR
$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$	Middle value of ordered data	Most frequent value	$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$	$\sigma = \sqrt{\sigma^2}$	IQR = $Q_3 - Q_1$

Skewness (g_1): $g_1 = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{s}\right)^3$

Kurtosis (g_2): $g_2 = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{s}\right)^4 - 3$

Where:
x_i = i-th observation
$\bar{x}$ = mean
n = number of observations
s = standard deviation
Q_1, Q_3 = first and third quartiles

KEY TAKEAWAY

Descriptive statistics summarize the central tendency, variability, distribution shape, and categorical composition of data, providing an interpretable numerical foundation for Exploratory Data Analysis and Explainable AI.

FROM RAW DATA TO EXPLAINABLE AI

Raw Data → Descriptive Statistics (EDA) → Understand Data Better → Build Better Models → Explainable Models → Trustworthy Decisions

Graphical EDA: Univariate Analysis

- **Histograms:** show the distribution shape of a single numeric variable.
- **Box plots:** summarize median, quartiles, and outliers compactly.
- **Bar charts:** display frequency of categorical variable levels.
- **Density plots (KDE):** smoothed estimate of a variable's probability distribution.



Graphical EDA: Multivariate Analysis

- **Scatter plots:** reveal relationships between two numeric variables.
- **Correlation heatmaps:** visualize pairwise correlation across many features at once.
- **Pair plots:** grid of scatter plots for all variable pairs.
- **Parallel coordinates plots:** compare multiple variables across observations simultaneously.



GRAPHICAL EDA FOR UNIVARIATE ANALYSIS

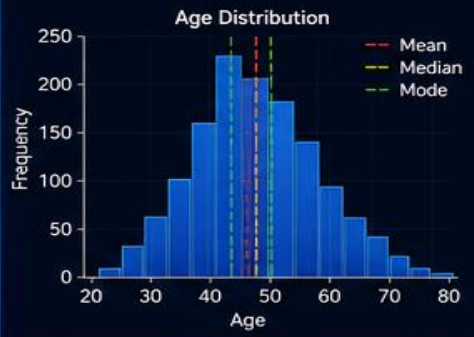
Visualizing the Distribution of a Single Variable

Univariate analysis examines one variable at a time to understand its distribution, central tendency, spread, and shape.

1. HISTOGRAM

Shows the frequency distribution of numerical data by dividing the range into bins.

Example: Distribution of Age



Summary Statistics						
Count	Mean	Median	Mode	Std Dev	Min	Max
1,000	44.7	45.0	42	15.6	18	80

Key Insights

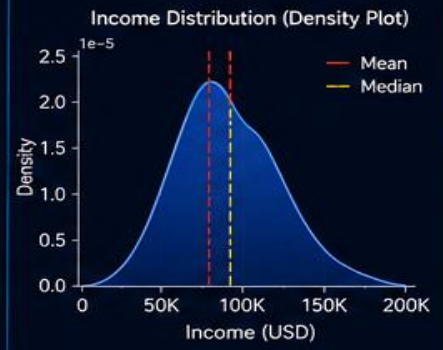
- Mean (red) slightly > Median (yellow) → Right-skewed distribution.
- Most ages lie between 30 and 60.

Best for: Understanding overall distribution and central tendency.

2. DENSITY PLOT

Smoothed version of histogram using a kernel density estimate.

Example: Distribution of Income



Key Insights

- Unimodal distribution.
- Slight right skew with a long right tail.

Best for: Identifying distribution shape and modality.

3. BOX PLOT

Summarizes distribution using five-number summary and identifies outliers.

Example: Distribution of Salary



Five-Number Summary				
Min	Q1 (25%)	Median (50%)	Q3 (75%)	Max
15,000	42,000	68,000	97,000	150,000

Key Insights

- Central 50% of salaries lie between 42K and 97K.
- Outliers present on both ends.

Best for: Detecting outliers and comparing spread.

4. VIOLIN PLOT

Combines box plot and density plot to show distribution shape and summary statistics.

Example: Distribution of Exam Scores



Key Insights

- Wider sections indicate higher density.
- Scores concentrated around 50–70.

Best for: Understanding distribution shape and density.

5. ECDF (EMPIRICAL CDF)

Shows the cumulative proportion of observations.

Example: ECDF of Waiting Time



Key Insights

- 50% of people wait less than 25 minutes (Median).
- 75% wait less than 50 minutes.

Best for: Understanding cumulative behavior and percentiles.

COMMON STATISTICS INTERPRETED THROUGH GRAPHS

Mean (Avg. Value)	Median (Middle Value)	Mode (Most Frequent Value)	Variance (Measure of Spread)	Std Dev (Spread in Original Units)
$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$			$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$	$\sigma = \sqrt{\sigma^2}$

WHY GRAPHICAL EDA?

- Quickly understand distribution shape
- Detect outliers and anomalies
- Identify skewness and modality
- Support data-driven decisions

KEY TAKEAWAY

Graphical univariate analysis provides an immediate and intuitive understanding of the distribution, helping to build better features and more explainable AI models.

Raw Data → Univariate Graphical EDA → Insights & Understanding → Better Features & Preprocessing → Explainable AI Models → Trustworthy Decisions

GRAPHICAL EDA FOR MULTIVARIATE ANALYSIS

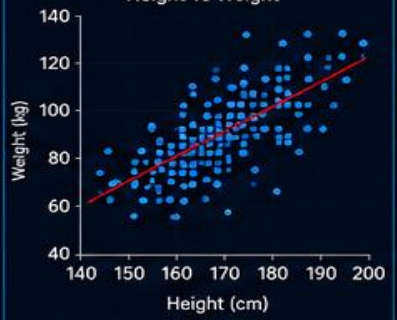
Visualizing Relationships, Patterns and Structure Across Multiple Variables

Multivariate analysis explores relationships between variables to uncover structure, correlations and patterns.

1. SCATTER PLOT

Visualizes the relationship between two continuous variables.

Example: Height vs Weight



Key Insights

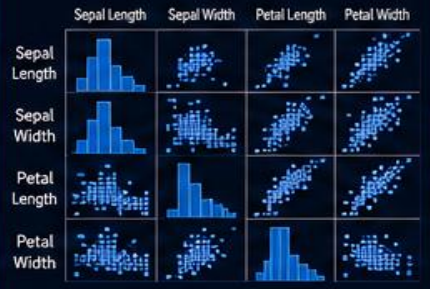
- Positive correlation between height and weight.
- Some high-weight outliers.

Best for: Exploring relationships between two variables.

2. PAIR PLOT (SCATTER MATRIX)

Displays scatter plots for all pairwise combinations in the dataset.

Example: Iris Dataset (4 Features)



Key Insights

- Petal length and petal width are strongly correlated.
- Sepal width has weak correlation with other features.

Best for: Understanding pairwise relationships across many variables.

3. HEATMAP (CORRELATION MATRIX)

Shows correlation strength and direction between variables.

Example: Correlation of Features



Key Insights

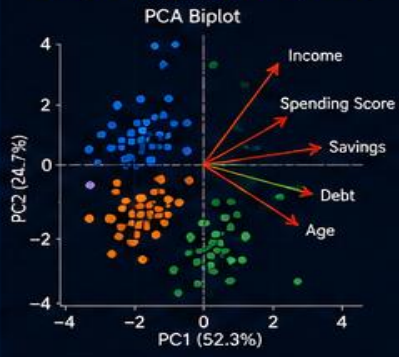
- Income and Spending Score are strongly positively correlated (0.62).
- Debt has negative correlation with Spending Score (-0.25).

Best for: Identifying linear relationships and multicollinearity.

4. BIPLOT (PCA)

Displays both observations and variables in reduced dimensions.

Example: PCA Biplot (2 Components)



Key Insights

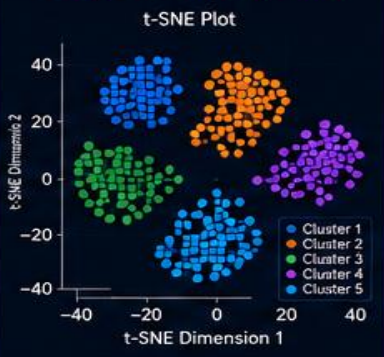
- PC1 mainly represents Income and Spending Score.
- Variables pointing in similar directions are positively related.

Best for: Understanding variable relationships in lower dimensions.

5. T-SNE / UMAP PLOT

Projects high-dimensional data to 2D/3D while preserving local structure.

Example: t-SNE (2D Projections)



Key Insights

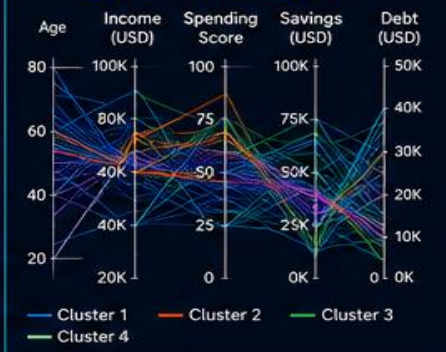
- Clear cluster separation in low-dimensional space.
- Useful for identifying groups and anomalies.

Best for: Discovering complex structures and clusters.

6. PARALLEL COORDINATES PLOT

Visualizes multivariate data across many dimensions.

Example: Customer Segments



Key Insights

- Cluster patterns across multiple dimensions.
- High income with high savings and low debt (Cluster 1).

Best for: Diomparing multivariate profiles across observations.

COMMON STATISTICS FOR MULTIVARIATE ANALYSIS

Correlation Coefficient (Pearson)

$$r_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

$$-1 \le r_{xy} \le 1$$

$r > 0$: Positive correlation
$r < 0$: Negative correlation

Covariance

$$\mathrm{Cov}(X,Y) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

Measures the direction of the linear relationship between variables.

Variance–Covariance Matrix

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1p} \\ \sigma_{21} & \sigma_2^2 & \cdots & \sigma_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{p1} & \sigma_{p2} & \cdots & \sigma_p^2 \end{bmatrix}$$

Shows variances on diagonals and covariances off-diagonals.

APPLICATIONS IN EXPLAINABLE AI (XAI)

- Identify important relationships and interactions between features.
- Detect multicollinearity and redundant features.
- Reveal hidden structure and clusters in data.
- Support feature selection and dimensionality reduction.
- Improve model interpretability and provide insights for domain experts.

KEY TAKEAWAY

Graphical multivariate analysis provides a powerful view of relationships and patterns across multiple variables, helping uncover hidden insights and enabling more **interpretable, trustworthy and explainable AI models.**

Raw Data → Multivariate Graphical EDA → Insights & Understanding → Better Features & Feature Engineering → Explainable AI Models → Trustworthy Decisions

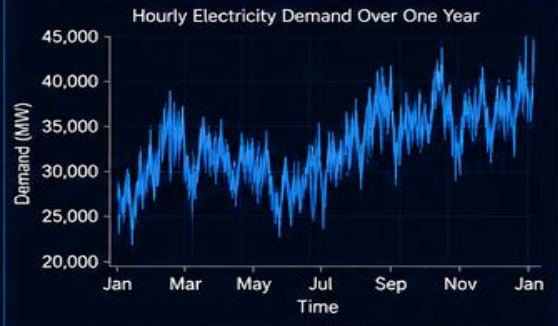
EDA and Time Series Data

- Time series EDA examines trend, seasonality, and noise components.
- Key plots: line plots over time, autocorrelation (ACF) and partial autocorrelation (PACF) plots, seasonal decomposition plots.
- Stationarity checks (e.g., Augmented Dickey-Fuller test) are essential before modeling.
- Understanding temporal structure aids later explanation of forecasting model behavior.

EDA FOR TIME SERIES DATA

Understanding Temporal Patterns Before Time Series Modeling

1. TIME SERIES LINE PLOT



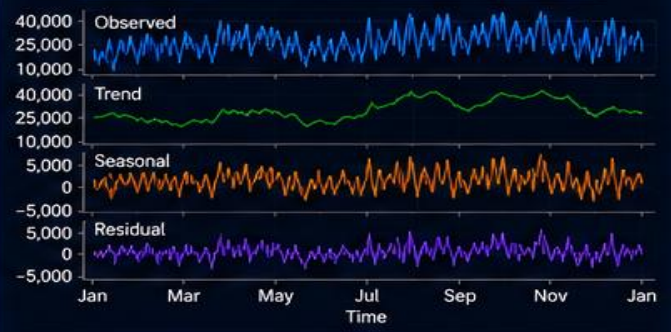
Visualize temporal patterns, trend and anomalies.

2. TREND ANALYSIS



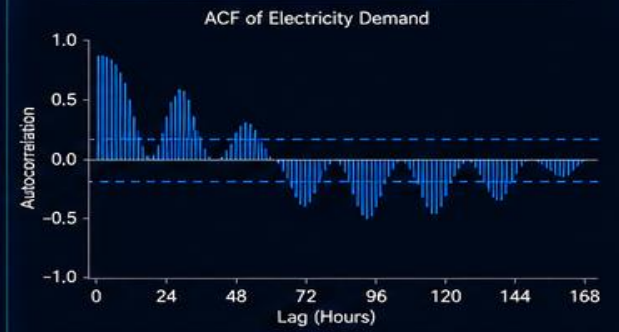
Long-term increasing/decreasing behavior.

3. SEASONAL DECOMPOSITION (STL)



Separates trend, seasonality and residuals.

4. AUTOCORRELATION (ACF)



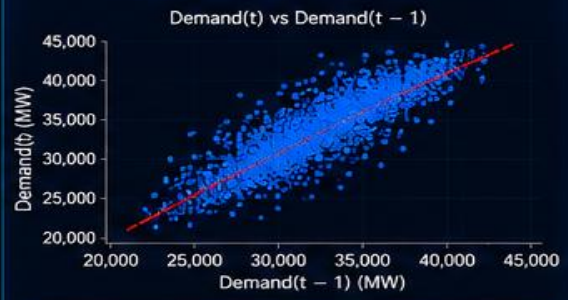
Measures correlation with previous lags.

5. PARTIAL AUTOCORRELATION (PACF)



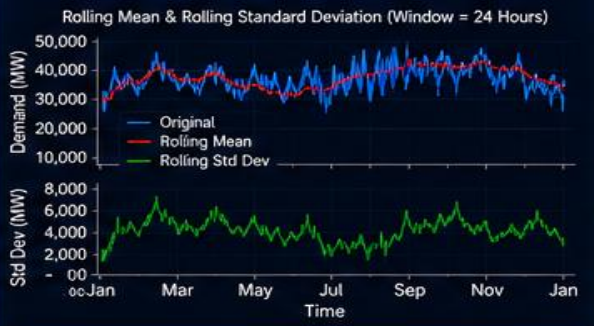
Identifies direct lag dependencies.

6. LAG PLOT (Lag = 1)



Detects serial correlation and nonlinear relationships.

7. ROLLING STATISTICS



Assesls stationarity over time.

8. DISTRIBUTION OVER TIME



Compare distributions across seasons and detect anomalies.

KEY TIME SERIES STATISTICS

Mean

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Average value of the series.

Rolling Mean (Window = w)

$$\mu_t = \frac{1}{w}\sum_{i=0}^{w-1} x^{-i}$$

Local average at time t.

Autocorrelation at Lag k

$$\rho_k = \frac{\mathrm{Cov}(X_t, X_{t-k})}{\sigma^2}$$

Correlation with value k steps back.

Variance

$$\sigma^2 = \frac{1}{n-1}\sum_{i=1}^{n} (x_i - \bar{x})^2$$

Measure of variability.

Differencing (Lag = 1)

$$\Delta X_t = X_t - X_{t-1}$$

Helps achieve stationarity.

WHY IT MATTERS?

- Reveals temporal patterns, seasonality and trends.
- Helps detect non-stationarity, outliers and structural breaks.
- Guides feature engineering and model selection.
- Improves interpretability and reliability of forecasts.



Raw Time Series → Visualization → Trend Analysis → Seasonality Analysis → Stationarity Check → Feature Engineering → Forecasting Models → Explainable AI

EDA and Natural Language Processing (NLP)

- Text-specific EDA: word frequency counts, word clouds, n-gram analysis.
- Document-level statistics: text length distribution, vocabulary size, sentiment distribution.
- Helps reveal class imbalance, vocabulary sparsity, and potential bias in text corpora before training.



EDA for Natural Language Processing (NLP)

Exploring and Understanding Text Data Before NLP Modeling

1. CORPUS STATISTICS

Understand the overall characteristics of the text corpus.

# Documents	12,500
# Sentences	215,230
# Tokens (Words)	3,42,512
Vocabulary Size	28,450
Avg. Sentence Length	18.7
Avg. Document Length	274.1

Language Distribution



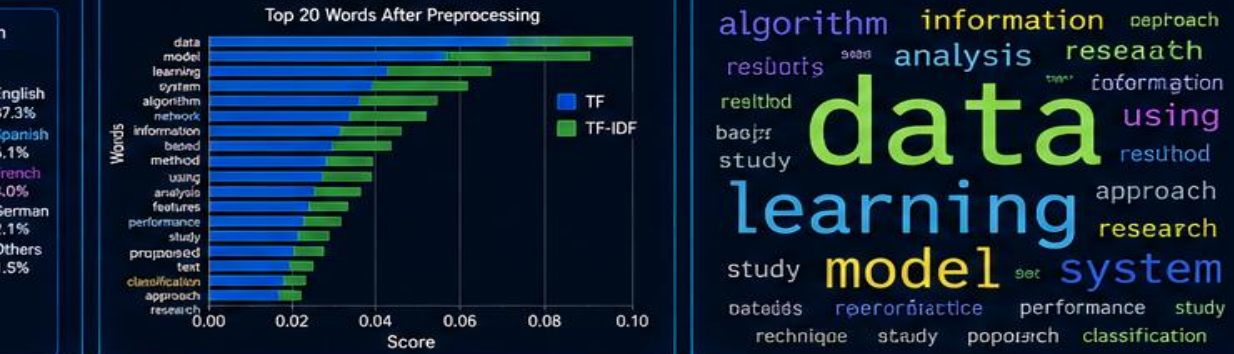
2. WORD FREQUENCY ANALYSIS

Identify frequently occurring and informative terms.

Top 20 Words After Preprocessing

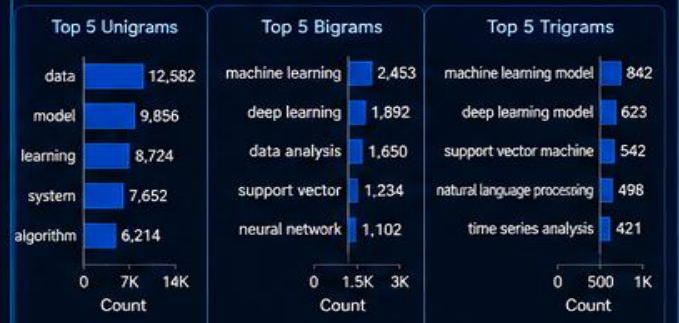
3. WORD CLOUD

Visualize dominant themes and concepts.



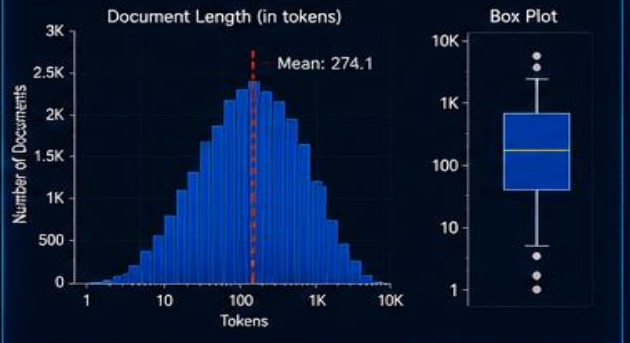
4. N-GRAM ANALYSIS

Discover common word sequences and contextual patterns.



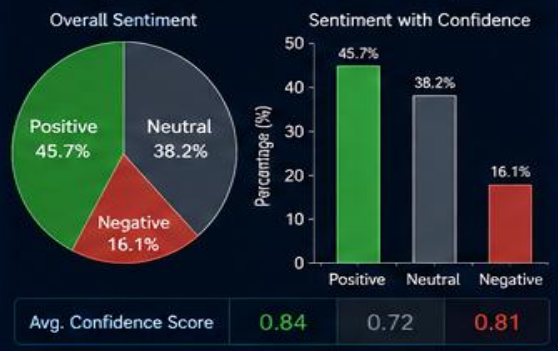
5. DOCUMENT LENGTH DISTRIBUTION

Detect unusually short or long documents.



6. SENTIMENT DISTRIBUTION

Analyze emotional distribution in the corpus.

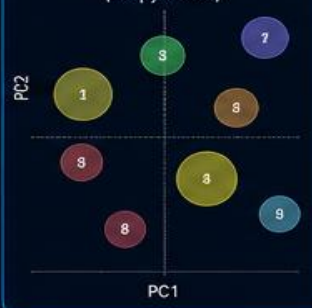


Avg. Confidence Score	0.84	0.72	0.81

7. TOPIC MODELING (LDA)

Identify hidden semantic topics within documents.

Intertopic Distance Map (via pyLDAvis)



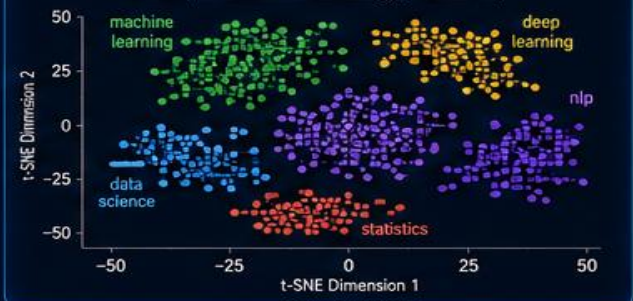
Top Words per Topic

- **Topic 1 – Machine Learning**: model, learning, algorithm, data, training, models, performance
- **Topic 2 – Natural Language Processing**: language, text, word, semantic, processing, corpus, information
- **Topic 3 – Computer Vision**: image, classification, object, detection, visual, recognition
- ...
- **Topic 10 – Time Series Analysis**: time, series, data, forecasting, analysis, model, prediction

8. WORD EMBEDDING VISUALIZATION

Visualize semantic similarity among words and documents.

t-SNE plot of Word Embeddings (Word2Vec)



KEY NLP STATISTICS & MEASURES

Term Frequency (TF)

$$\mathrm{TF}(t) = f(t,d)$$

$f(t,d)$ = frequency of term t in document d

Inverse Document Frequency (IDF)

$$\mathrm{IDF}(t) = \log\left(\frac{N}{df(t)}\right)$$

N = total number of documents
$df(t)$ = number of documents containing term t

TF–IDF

$$\mathrm{TF\text{–}IDF}(t,d) = \mathrm{TF}(t,d) \times \mathrm{IDF}(t)$$

Measures the importance of term t in document d relative to corpus

Cosine Similarity

$$\mathrm{sim}(A,B) = \frac{A \cdot B}{\|A\|\,\|B\|}$$

$A \cdot B$ = dot product of vectors
$\|A\|, \|B\|$ = vector magnitudes

Perplexity (Language Models)

$$\mathrm{Perplexity} = 2^{-\frac{1}{N}\sum_{i=1}^{N}\log_2 P(w_i)}$$

Lower perplexity indicates better model performance

Lexical Diversity

$$\mathrm{LD} = \frac{\text{Unique Tokens}}{\text{Total Tokens}}$$

Measures richness of vocabulary

Raw Text → Cleaning & Tokenization → Text Statistics → Word Frequency → Topic Discovery → Feature Extraction → NLP Models → Explainable AI

EDA and Computer Vision

- Image-specific EDA: class distribution across image categories, image size/aspect ratio statistics.
- Pixel intensity histograms and channel-wise statistics (RGB).
- Visual inspection of sample images per class to detect labeling errors or dataset bias.
- Detecting duplicate or near-duplicate images that could cause data leakage.



EDA AND COMPUTER VISION

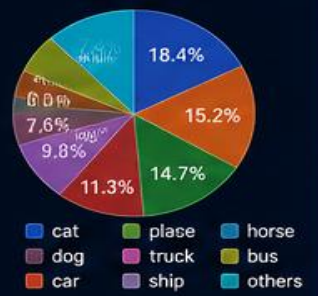
Exploring and Understanding Image Data Before Building CV Models

1. DATASET OVERVIEW

Understand the composition and balance of the dataset.

- Total Images: 25,632
- Classes: 10
- Image Size (Avg.): 256 × 256
- Color Channels: 3 (RGB)
- File Format: .jpg / .png
- Total Size: 12.6 GB

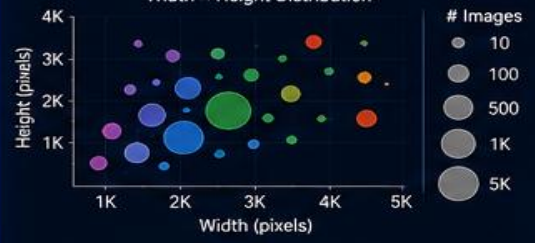
Class Distribution



2. IMAGE RESOLUTION ANALYSIS

Analyze the distribution of image resolutions.

Width × Height Distribution



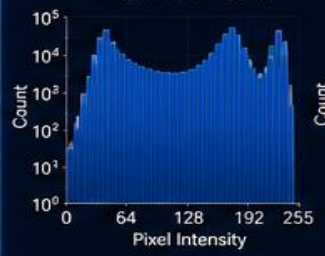
Most Common Resolutions

224×224	256×256	512×512	640×480	1024×768
6,812	5,421	4,231	2,103	1,876

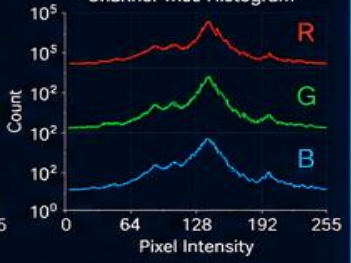
3. PIXEL INTENSITY ANALYSIS

Examine pixel value distribution.

Grayscale Histogram



Channel-wise Histogram



Summary Statistics (All Channels)

Mean	Std. Dev	Min	Max	Skewness	Kurtosis
127.6	73.8	0	255	0.12	2.71

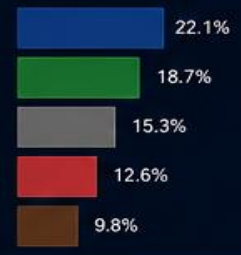
4. COLOR ANALYSIS

Understand color distribution and dominance.

Color Space Histogram (HSV)



Dominant Colors (Top 5)



Average Color (RGB)

R: 125 G: 132 B: 137

5. DATA AUGMENTATION PREVIEW

Visualize the effect of common augmentations.

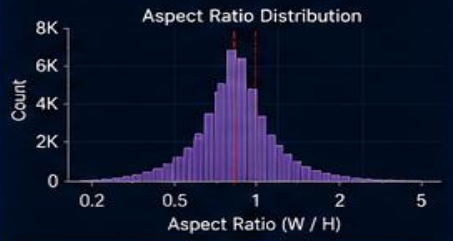


Helps in understanding robustness and variability.

6. ASPECT RATIO ANALYSIS

Analyze the distribution of width-to-height ratios.

Aspect Ratio Distribution



Summary

Min	Q1	Median	Q3	Max
0.21	0.75	1.15	1.78	4.92

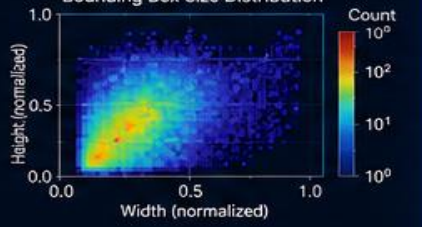
7. LABEL ANALYSIS

Check label quality and potential issues.

Label Count per Image



Bounding Box Size Distribution



Annotation Issues

Empty Images	Small Objects (area < 1%)	Partially Visible	Duplicate Images
1,243 (4.8%)	2,105 (8.2%)	3,412 (13.3%)	892 (3.5%)

8. CORRELATION BETWEEN FEATURES

Explore relationships between numerical features.

	Width	Height	Area	Bright.	Contrast	Satur.
Width	1.00	0.78	0.92	-0.12	0.05	0.01
Height	0.78	1.00	0.94	-0.10	0.03	-0.02
Area	0.92	0.94	1.00	-0.11	0.02	-0.03
Brightness	-0.12	-0.10	-0.11	1.00	0.62	0.58
Contrast	0.05	0.03	0.02	0.62	1.00	0.71
Saturation	0.01	-0.02	-0.03	0.58	0.71	1.00

Helps identify redundant or highly correlated features.

KEY STATISTICS & FORMULAS

Mean Pixel Intensity (per channel)

$$\mu_c = \frac{1}{N}\sum_{i=1}^{N} P_{i,c}$$

Average pixel value for channel c.

Standard Deviation

$$\sigma_c = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(P_{i,c}-\mu_c)^2}$$

Spread of pixel values.

Image Entropy

$$H = -\sum_{i=0}^{255} p(i)\log_2 p(i)$$

Measures information content.

Sparsity (for bboxes)

$$S = 1 - \frac{\sum_{k=1}^{K} A_k}{W \times H}$$

K: # boxes, A_k: area of kth box

Aspect Ratio

$$AR = \frac{Width}{Height}$$

Shape descriptor.

Brightness

$$B = 0.299R + 0.587G + 0.114B$$

Perceived brightness (luma).

WHY EDA FOR COMPUTER VISION?

- Understand data characteristics & quality
- Detect class imbalance & annotation issues
- Guide data augmentation & preprocessing
- Improve model performance & generalization
- Build more explainable & trustworthy CV models

Raw Image Data → Data Cleaning & Filtering → Exploratory Data Analysis → Feature & Label Insights → Data Augmentation Strategy → Feature Extraction → CV Models (Training) → Explainable AI (XAI)

Feature Engineering: Overview

- Feature engineering is the process of using domain knowledge to create, transform, or select input variables that improve model performance and interpretability.
- Well-engineered features can make even simple, transparent models highly competitive with complex black boxes.
- Includes encoding, scaling, binning, interaction terms, and dimensionality reduction.



Feature Engineering and Explainability

- Meaningful, human-understandable features directly translate into meaningful explanations (e.g., "age" is more interpretable than an opaque PCA component).
- Avoiding overly engineered or abstract features (e.g., deep embeddings) preserves interpretability, at some potential cost to accuracy.
- Feature selection (removing irrelevant/redundant variables) simplifies both the model and its later explanations.

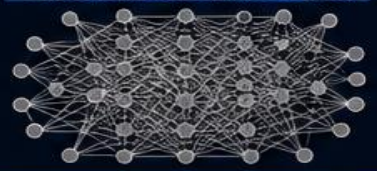
ROLE OF TRADITIONAL FEATURE ENGINEERING IN EXPLAINABLE AI (XAI)

How Feature Engineering Improves Model Interpretability and Trust

1. REDUCE MODEL COMPLEXITY

Fewer features reduce model complexity, improve interpretability, and eliminate redundant information.

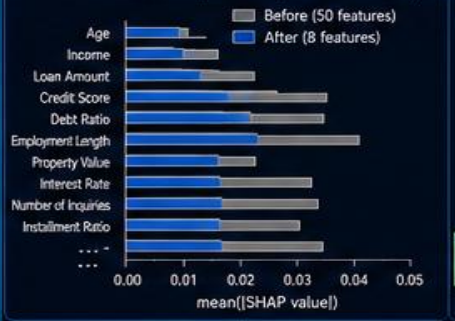
Before Feature Selection (50 Features)



After Feature Selection (8 Features)



SHAP Feature Importance (Before vs After)



Decision Tree (After Feature Selection)



Feature selection and reduction simplify the model, reduce interactions, remove noise and make output more predictable and interpretable.

2. IMPROVE COMPUTATIONAL EFFICIENCY

Smaller feature sets reduce storage, accelerate computation, and simplify automated model optimization.

Metric	Before Feature Engineering (50 Features)	After Feature Engineering (8 Features)	Improvement
Storage (Dataset Size)	1.82 GB	0.32 GB	82.4% ↓
Memory Usage (Training)	3.45 GB	0.78 GB	77.4% ↓
Training Time (Per Epoch)	14.6 min	3.1 min	78.8% ↓
Inference Latency (Per Prediction)	58.4 ms	12.7 ms	78.3% ↓
CPU Utilization (Training)	86%	32%	62.8% ↓

Automated Model Optimization

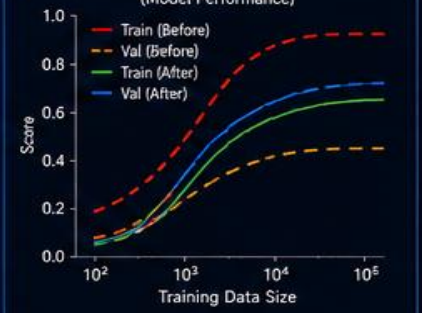


Lower cost, faster computation, and easier model building across hyperparameters and algorithms.

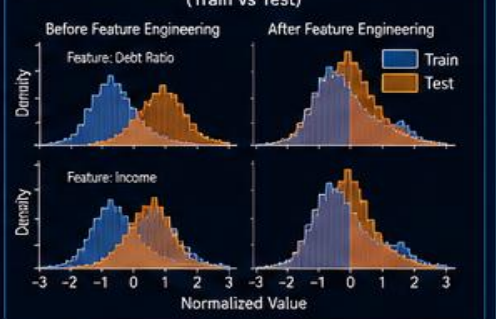
3. REDUCE OVERFITTING

Feature analysis minimizes overfitting, improves generalization, and increases reliability on unseen data.

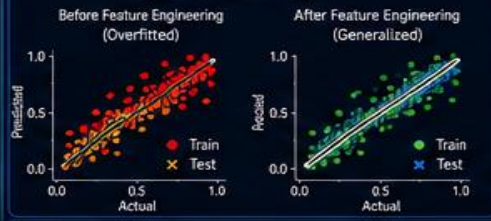
Learning Curves (Model Performance)



Feature Distribution Shift (Train vs Test)



Overfitting vs Generalization (Prediction Performance)



Generalization Metrics

Metric	Before	After	Improvement
Train R^2	0.97	0.89	–
Test R^2	0.62	0.88	+41.9%
RMSE (Test)	0.286	0.142	50.3% ↓
MAE (Test)	0.213	0.098	54.0% ↓

Understanding feature distributions and removing irrelevant/noisy features improves generalization and model reliability.

Raw Data → Feature Selection → Feature Engineering → Simpler Model → Better Generalization → Explainable AI (XAI)

BENEFITS OF FEATURE ENGINEERING

IMPROVED INTERPRETABILITY
Simpler models with meaningful features are easier to explain and understand.

REDUCED NOISE
Removes irrelevant and redundant information, improving signal and model clarity.

FASTER COMPUTATION
Lower storage, less memory usage, and shorter training and inference time.

BETTER PREDICTION ACCURACY
Improved generalization and robustness on unseen data.

KEY STATISTICS & FORMULAS

Information Gain

$$IG = H(Y) - H(Y|X)$$

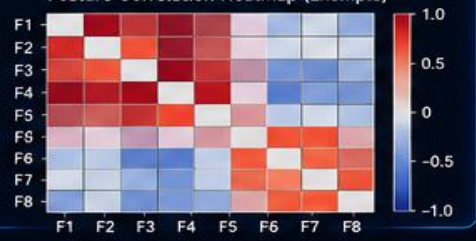
Measures the reduction in entropy of target variable Y by knowing X.

Variance Threshold

$$Var(X) > \tau$$

Remove features whose variance is below threshold τ.

Feature Correlation Heatmap (Example)



> Good features lead to simple models. Simple models lead to explainability. Explainability leads to trust.

Feature Engineering: Taxonomy and Tools

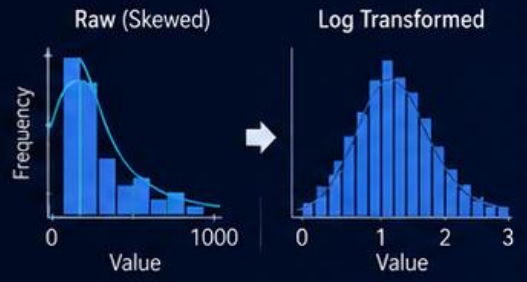
- **Transformation:** scaling, normalization, log/power transforms.
- **Encoding:** one-hot encoding, ordinal encoding, target encoding for categorical variables.
- **Construction:** interaction terms, polynomial features, domain-driven ratios.
- **Selection / Reduction:** filter, wrapper, embedded methods; PCA, t-SNE for dimensionality reduction.
- **Tools:** scikit-learn, Featuretools, pandas, category_encoders.

FEATURE ENGINEERING TECHNIQUES

Transforming Raw Data into Meaningful Features for Better AI Models

1. TRANSFORMATION

- Scaling (Min-Max, Standard)
- Normalization
- Log Transform
- Power Transform



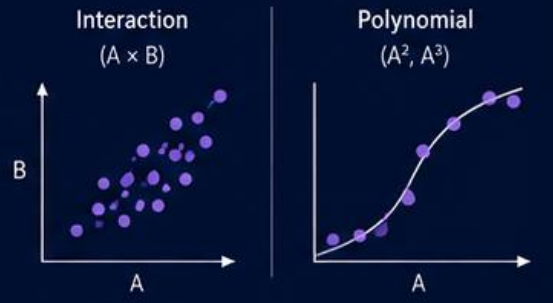
2. ENCODING

- One-Hot Encoding
- Ordinal Encoding
- Target Encoding

Category	One-Hot Encoding
Red	[1, 0, 0]
Green	[0, 1, 0]
Blue	[0, 0, 1]

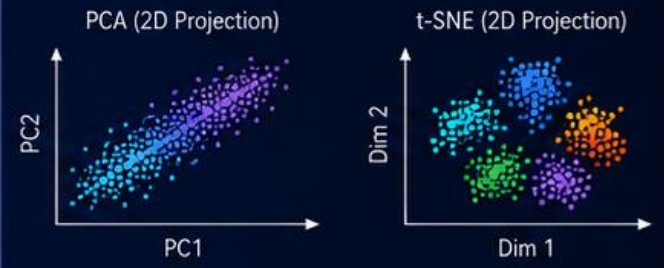
3. FEATURE CONSTRUCTION

- Interaction Terms
- Polynomial Features
- Domain-driven Ratios



4. FEATURE SELECTION / REDUCTION

- Filter Methods
- Wrapper Methods
- Embedded Methods
- PCA
- t-SNE



5. TOOLS & LIBRARIES

scikit-learn | pandas | Featuretools | category_encoders



Effective feature engineering **improves model accuracy**, **reduces complexity**, **accelerates training**, and **enhances explainability**.

FEATURE ENGINEERING – DATA TRANSFORMATION

Transforming Raw Features for Better Model Performance

RAW DATA
Different scales, skewed distributions, outliers, units

→

TRANSFORMATION
Scaling / Normalization / Log / Power

→

MACHINE LEARNING MODEL
Improved convergence, higher accuracy, better generalization

1 SCALING (STANDARDIZATION)

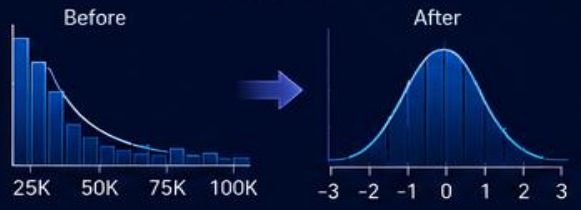
$$z = \frac{x - \mu}{\sigma}$$

Centers data around 0 with unit variance

Income (₹)	Scaled Value (z)
25,000	−1.32
40,000	−0.62
55,000	0.08
80,000	1.10
100,000	1.76



Distribution

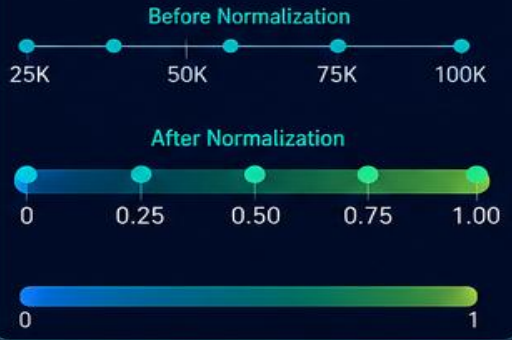


2 NORMALIZATION (MIN–MAX SCALING)

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Scales data to a fixed range [0, 1]

Income (₹)	Normalized Value (x')
25,000	0.00
40,000	0.20
55,000	0.40
80,000	0.73
100,000	1.00



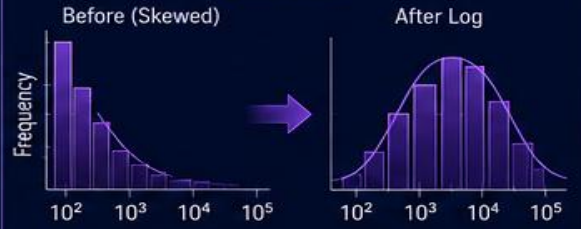
3 LOG TRANSFORMATION

$$x' = \log(x + 1)$$

Reduces right skewness, compresses large values

Amount (₹)	Log Value
100	4.61
500	6.22
1,000	6.91
10,000	9.21
100,000	11.51

Distribution



Makes data more symmetric and easier for models to learn

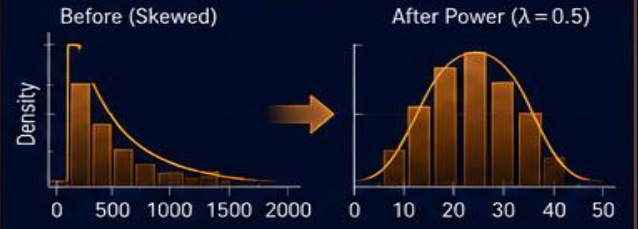
4 POWER TRANSFORMATION (BOX–COX / YEO–JOHNSON)

$$y = x^{\lambda}$$

Stabilizes variance and reduces skewness

x (Original)	Power (λ = 0.5) ($\sqrt{x}$)
25	5
100	10
400	20
900	30
1,600	40

Distribution



Transforms data towards normality and stabilizes variance

SCALING
Standardization
Mean = 0
Std Dev = 1

NORMALIZATION
Range 0 – 1
Min = 0
Max = 1

LOG TRANSFORM
Reduce Skewness
Compress large values
Handle outliers

POWER TRANSFORM
Stabilize Variance
Improve symmetry
Better model assumptions

FEATURE ENGINEERING – ENCODING CATEGORICAL VARIABLES

Convert categorical variables into numerical format for machine learning models

Raw Data — Categorical Variables → **Encoding** — Convert to Numerical → **Machine Learning** — Model Training & Prediction

1 ONE-HOT ENCODING

Create a new binary column for each category.

Example: Customer Churn Dataset

ID	Gender	City
1	Male	Delhi
2	Female	Mumbai
3	Male	Bangalore
4	Female	Delhi
5	Male	Mumbai

One-Hot Encoded Output

Gender_Male	Gender_Female
1	0
0	1
1	0
0	1
1	0

- Best for nominal (no order) categories
- Increases dimensionality

WHEN TO USE

- When categories have no natural order.

EXAMPLE

Gender, City, Product Type

2 ORDINAL ENCODING

Map categories based on natural order.

Example: Education Level

ID	Education Level	Income (₹)
1	High School	30,000
2	Diploma	40,000
3	Graduate	60,000
4	Postgraduate	80,000
5	PhD	1,20,000

Ordinal Encoded Output

Education_Level_Ordinal
1
2
3
4
5

- Preserves the ordinal relationship between categories
- Suitable for ordinal features

WHEN TO USE

- When categories have a meaningful order.

EXAMPLE

Education Level, T-Shirt Size (S < M < L < XL)

Category → Code (Example)	
High School	→ 1
Diploma	→ 2
Graduate	→ 3
Postgraduate	→ 4
PhD	→ 5

3 TARGET ENCODING

Replace category with the target mean.

Example: City vs Churn (Target = Churn)

ID	City	Churn
1	Delhi	1
2	Mumbai	0
3	Bangalore	1
4	Delhi	0
5	Mumbai	0
6	Bangalore	1
7	Chennai	0

Target Encoded Output (Mean of Churn)

City_Target_Enc
0.50
0.00
1.00
0.50
0.00
1.00
0.00

- Replaces category with target (mean) value
- Reduces dimensionality | Risk of target leakage

HOW TARGET ENCODING WORKS (City Example)

City	Sum of Churn	Count	Mean Target (Churn)
Delhi	1	2	1/2 = 0.50
Mumbai	0	2	0/2 = 0.00
Bangalore	2	2	2/2 = 1.00
Chennai	0	1	0/1 = 0.00

Important:
Use cross-validation or smoothing to avoid overfitting and target leakage.

COMPARISON SUMMARY

One-Hot Encoding	Ordinal Encoding	Target Encoding
✓ No assumption about order	✓ Preserves order information	✓ Low dimensionality
✓ Works well for tree models	✓ Low dimensionality	✓ Captures target relationship
✗ High dimensionality	✗ Not suitable for nominal data	✗ Risk of overfitting / leakage

KEY TAKEAWAY

Choose the encoding method based on the nature of the categorical variable and the model you are building.

FEATURE ENGINEERING – CONSTRUCTION

Create new features by combining existing ones to capture hidden patterns

Raw Features
Basic features from data

→

Construct New Features
Interaction Terms / Polynomial Features / Domain Ratios

→

Better Representations
Improved predictive power and model performance

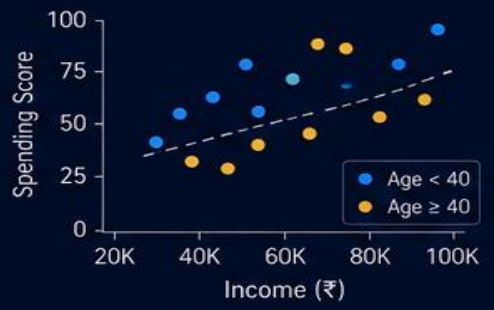
1 INTERACTION TERMS

Capture the combined effect of two or more features.

Example: Customer Purchase Data

Customer_ID	Age	Income (₹)	Spending Score	Age × Income
1	25	40,000	65	1,000,000
2	35	70,000	80	2,450,000
3	45	90,000	90	4,050,000
4	55	60,000	70	3,300,000
5	65	50,000	60	3,250,000

Why Interaction Helps?



At similar income levels, older customers tend to spend more.

Interaction term (Age × Income) helps the model learn this effect.

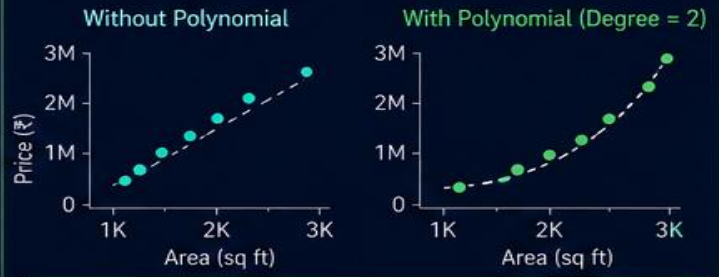
2 POLYNOMIAL FEATURES

Capture non-linear relationships by adding powers of features.

Example: House Price Prediction

Area (sq ft)	Area²	Area³
1000	1,000,000	1,000,000,000
1500	2,250,000	3,375,000,000
2000	4,000,000	8,000,000,000
2500	6,250,000	15,625,000,000
3000	9,000,000	27,000,000,000

Effect on Relationship (Area vs Price)



Polynomial features allow models like Linear Regression to learn curved (non-linear) relationships.

3 DOMAIN-DRIVEN RATIOS

Use domain knowledge to create meaningful ratios.

Example: E-commerce Customer Data

Customer_ID	Total Spent (₹)	Total Orders	Avg Order Value (₹)	Return Rate (%)
1	12,000	6	2,000.00	5.2
2	25,000	10	2,500.00	2.4
3	8,000	3	2,666.67	8.3
4	30,000	15	2,000.00	1.7
5	18,000	9	2,000.00	4.8

Common Domain Ratios

- Average Order Value = $\frac{\text{Total Spent}}{\text{Total Orders}}$
- Return Rate (%) = $\frac{\text{Total Returns}}{\text{Total Orders}} \times 100$
- Spend per Day = $\frac{\text{Total Spent}}{\text{Days Active}}$
- Discount Utilization = $\frac{\text{Discount Used}}{\text{Total Spent}} \times 100$

BENEFITS

- Capture complex patterns
- Improve model performance
- Make linear models more powerful
- Better generalization

SUMMARY COMPARISON

Method	Purpose	Example	Best Used With	Risk / Caution
Interaction Terms	Capture combined effects	Age × Income	Linear Models, Tree Models	High dimensionality
Polynomial Features	Model non-linear relationships	Area², Area³	Linear / GLM Models	Overfitting if degree is high
Domain-Driven Ratios	Add business meaning	Avg Order Value, Return Rate	All Models	Incorrect ratios can mislead

KEY TAKEAWAY

Well-constructed features provide more information to the model, leading to better predictions with fewer assumptions.

FEATURE ENGINEERING – SELECTION

Select the most relevant features and remove the rest

All Features
Many features may include noise and irrelevant information

→

Feature Selection
Filter / Wrapper / Embedded Methods

→

Selected Features
Reduced dimensionality, improved accuracy, faster training

1 FILTER METHODS

Select features based on statistical measures.
Model independent.

Example: Correlation with Target (|r|)

Feature	Correlation (r)	|r|	Rank
Age	0.72	0.72	1
Income	0.65	0.65	2
Credit Score	0.58	0.58	3
Tenure	0.22	0.22	4
Gender	0.05	0.05	5
City	0.01	0.01	6



Common Techniques
- Correlation, Chi-square
- Mutual Information
- Variance Threshold

Pros
- ✓ Fast and simple
- ✓ Works well for high dimensional data

Cons
- ✗ Ignores feature interaction
- ✗ May select redundant features

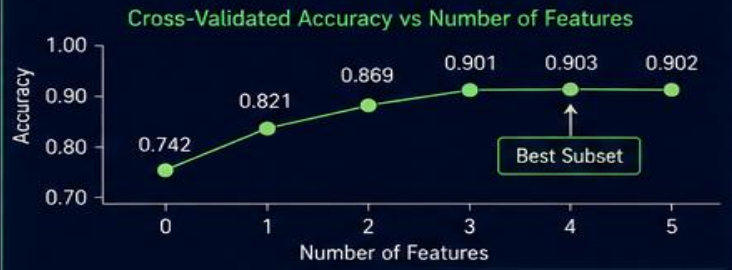
Best Used With: Very high dimensional datasets (text data, genomic data, etc.)

2 WRAPPER METHODS

Tries different subsets of features using a model.
Model dependent.

Example: Forward Selection with Logistic Regression

Step	Selected Features	CV Accuracy
0	—	0.742
1	Age	0.821
2	Age, Income	0.869
3	Age, Income, Credit Score	0.901
4	Age, Income, Credit Score, Tenure	0.903
5	Age, Income, Credit Score, Tenure, Gender	0.902



Pros
- ✓ Considers feature interaction
- ✓ Usually gives better accuracy

Cons
- ✗ Computationally expensive
- ✗ Not suitable for very high dimensional data

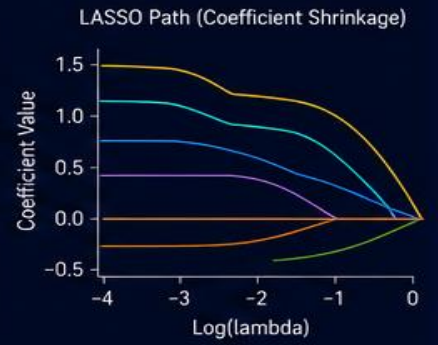
Best Used With: Small to medium feature sets where highest accuracy is critical

3 EMBEDDED METHODS

Feature selection occurs during model training.
Model dependent.

Example: LASSO (L1 Regularization)

Feature	Coefficient	Selected?
Age	1.245	Yes
Income	0.876	Yes
Credit Score	0.654	Yes
Tenure	0.000	No
Gender	0.000	No
City	0.000	No
Marital Status	0.412	Yes
Education	0.208	Yes



Common Techniques
- LASSO (L1)
- Ridge (L2)
- Elastic Net
- Tree-based (RF, XGBoost)

Pros
- ✓ Built-in to model training
- ✓ Handles high dimensional data well

Cons
- ✗ Less interpretable than filter
- ✗ May miss useful features if over-regularized

Best Used With: High dimensional data when using regularized models or tree-based models

METHOD COMPARISON SUMMARY

Method	Model Dependency	Speed	Captures Interaction	Best For
Filter	No	Very Fast	No	Large datasets
Wrapper	Yes	Slow	Yes	Small/Medium datasets
Embedded	Yes	Fast	Yes (partial)	Regularized / Tree models

TYPICAL WORKFLOW

Dataset with All Features → Choose a Selection Method → Select Top Features → Train Model → Better Accuracy & Efficiency

KEY TAKEAWAY

Feature selection improves model performance, reduces overfitting, lowers training time, and enhances interpretability.

PRINCIPAL COMPONENT ANALYSIS (PCA)

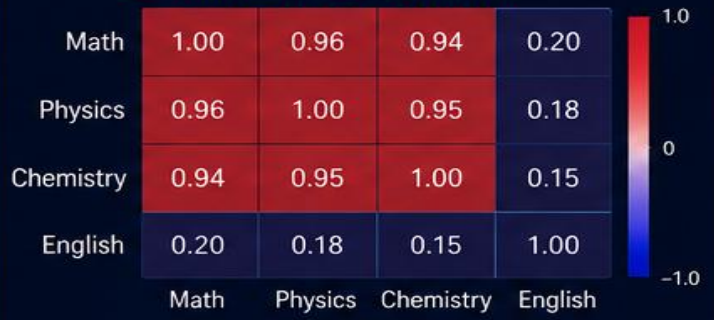
Reducing 4 Features into 2 Principal Components

1. ORIGINAL DATASET (4 FEATURES)

Student	Math (X_1)	Physics (X_2)	Chemistry (X_3)	English (X_4)
A	95	93	90	75
B	88	86	84	70
C	80	81	79	72
D	72	70	68	68
E	65	64	62	66

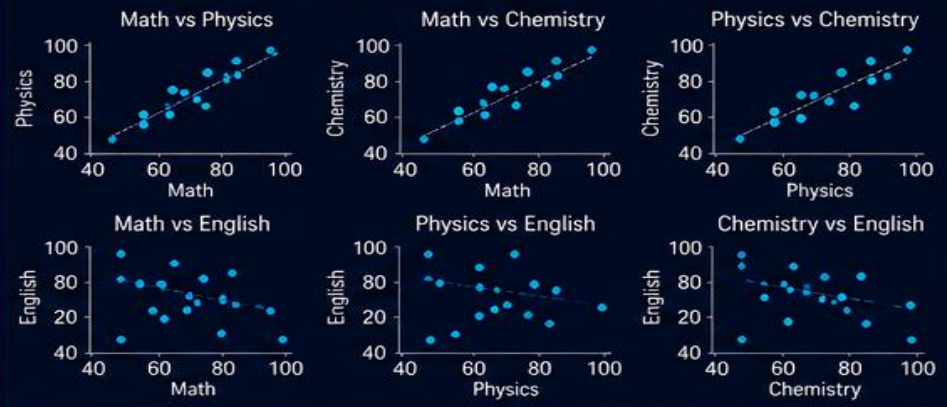
Math, Physics and Chemistry are highly correlated. English is different.

2. CORRELATION HEATMAP



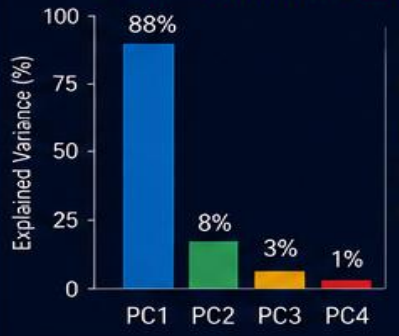
High correlation among Math, Physics and Chemistry less correlated.

3. DATA VISUALIZATION (PAIR PLOTS)



Science subjects move together, English behaves differently.

4. PCA – VARIANCE EXPLAINED



Cumulative Variance

PC1 = 88%

PC1 + PC2 = 96%

PC1 + PC3 = 99%

PC1 + PC4 = 100%

Keeping PC1 and PC2 preserves 96% of the information.

5. PRINCIPAL COMPONENT EXPRESSIONS

PC1	$= 0.58X_1 + 0.57X_2 + 0.56X_3 + 0.15X_4$
PC2	$= -0.14X_1 + 0.28X_2 - 0.23X_3 + 0.92X_4$
PC3	$= 0.66X_1 - 0.73X_2 + 0.17X_3 + 0.10X_4$
PC4	$= 0.45X_1 + 0.29X_2 + 0.78X_3 - 0.31X_4$

X_1 = Math, X_2 = Physics, X_3 = Chemistry, X_4 = English

Each PC is a linear combination of the original features.

PCs are orthogonal (uncorrelated with each other).

6. EIGENVALUES & VARIANCE CALCULATION

PC	Eigenvalue (λ)	Variance (%)	Calculation
PC1	3.52	88.00%	(3.52 / 4) × 100 = 88.00%
PC2	0.32	8.00%	(0.32 / 4) × 100 = 8.00%
PC3	0.12	3.00%	(0.12 / 4) × 100 = 3.00%
PC4	0.04	1.00%	(0.04 / 4) × 100 = 1.00%

Total Variance = 3.52 + 0.32 + 0.12 + 0.04 = 4.00 (= number of features)

Explained Variance (%) = (λ_i / Total Variance) × 100

7. DATA AFTER PCA (KEEPING PC1 & PC2)

Student	PC1	PC2
A	162.1	5.8
B	150.3	4.1
C	139.6	−1.2
D	122.4	−3.6
E	109.8	−5.1

Reduced from 4 Features to 2 Components

Only 2 components retained with ~96% information.

8. PCA PROCESS FLOW



9. WHY PCA?

- Reduces dimensionality
- Removes redundancy
- Speeds up ML models
- Reduces noise
- Improves visualization
- Helps in better model generalization

t-SNE (t-Distributed Stochastic Neighbor Embedding)

Visualizing High-Dimensional Data in 2D

Example: Student Performance Dataset (4 Features)

1 Original Dataset (4 Features)

Student	Math	Physics	Chemistry	English
A	95	93	90	75
B	88	86	84	70
C	80	81	79	72
D	72	70	68	68
E	65	64	62	66

ⓘ Each student is a point in 4-dimensional space.

2 Pairwise Euclidean Distance

	A	B	C	D	E
A	0	12.61	22.69	37.95	49.59
B	12.61	0	13.89	28.21	39.26
C	22.69	13.89	0	15.72	26.00
D	37.95	28.21	15.72	0	13.27
E	49.59	39.26	26.00	13.27	0

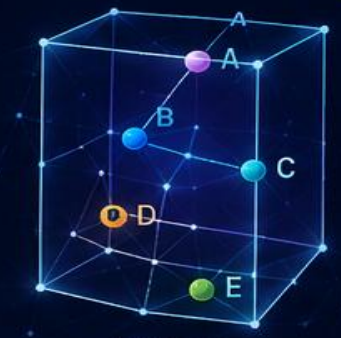
$$d(A,B) = \sqrt{(95-88)^2 + (93-86)^2 + (90-84)^2 + (75-70)^2} = 12.61$$

3 Convert Distance to Similarity (Gaussian)

$$P_{ij} = \frac{e^{-d_{ij}^2/2\sigma^2}}{\sum_k e^{-d_{ik}^2/2\sigma^2}}, \qquad \sigma = 20$$

Pair	w.r.t A	w.r.t B	w.r.t C	w.r.t D	w.r.t E
A–B	0.41	0.46	–	–	–
A–C	0.30	–	0.45	–	–
A–D	0.18	–	–	0.46	–
A–E	0.11	–	–	–	0.49

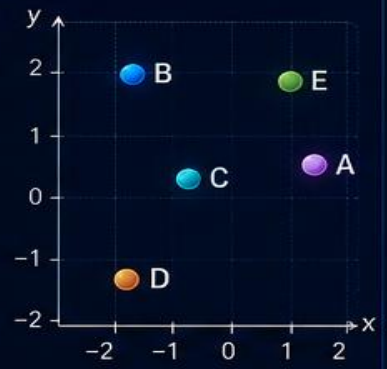
ⓘ Higher probability → more similar (closer)



4D Space

4 Initial 2D Random Configuration (Before Optimization)

Student	x (random)	y (random)
A	1.20	−0.80
B	−0.90	1.34
C	0.40	0.10
D	−1.10	−1.20
E	0.80	1.20



ⓘ t-SNE starts with random 2D positions.

5 KL Divergence Optimization

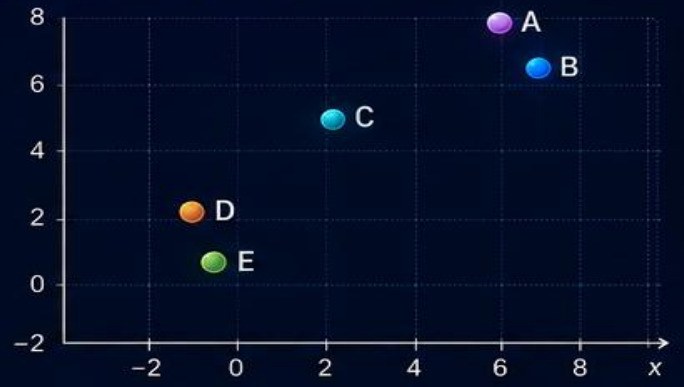
Objective: Minimize difference between high-D similarity P_{ij} and low-D similarity Q_{ij}

$$KL(P||Q) = \sum_{i,j} P_{ij} \log\left(\frac{P_{ij}}{Q_{ij}}\right)$$

- Adjust 2D positions iteratively
- Preserve local neighborhoods
- Similar points → stay close
- Dissimilar points → move apart



6 Final 2D t-SNE Visualization



- A & B: Very similar (High marks in all subjects)
- C: Moderate performance
- D & E: Lower performance group (similar)

ⓘ t-SNE reveals natural clusters based on similarity.

t-SNE Workflow

Input Data (High-D) → Pairwise Distances → Convert to Probabilities P_{ij} → Optimize (Minimize KL Divergence) → 2D Visualization (Clusters)

t-SNE vs PCA

Aspect	PCA	t-SNE
Goal	Reduce dimensions (linear)	Visualize clusters (non-linear)
Preserves	Global structure (variance)	Local structure (neighborhoods)
Output	New features (PC1, PC2, ...)	2D/3D coordinates for plotting
Use in ML	Yes (preprocessing)	No (visualization only)

Key Takeaway

t-SNE places similar students close together in 2D while keeping dissimilar students far apart, making hidden patterns easy to see.

Pitfalls in Feature Engineering (Skipping EDA)

Common mistakes that lead to poor models, misleading results and hard-to-explain AI

1 Skipping EDA and moving straight to modeling

Hides data quality issues that later masquerade as "model errors."

Why it's a problem

- Missing values, outliers, duplicates remain hidden
- Data leakage goes unnoticed
- Wrong patterns are learned

Example

Target = Loan Default (Yes/No)



Consequence

Poor model performance is wrongly blamed on "algorithm choice" instead of data issues.

2 Creating overly complex engineered features

That are not traceable back to domain meaning.

Why it's a problem

- Hard to interpret
- Brittle to data drift
- Difficult to validate and explain

Example

Hand-crafted Feature

$$F_{\text{complex}} = (x_1 * \log(x_3 + 1))^2 + \sqrt{x_2} \,/\, (x_4 + 0.0001) - {x_5}^3$$

vs. Simple & Meaningful Features

Raw Feature	Meaning
x_1: Monthly Income	Ability to repay
x_2: Existing EMI	Debt burden
x_3: Credit Score	Creditworthiness
x_4: Loan Tenure	Commitment
x_5: No. of Inquiries	Risk signal

Consequence

Models become opaque and fail regulatory / domain audits.

3 Ignoring class imbalance or sampling bias

discovered during EDA.

Why it's a problem

- Model biased toward majority class
- Poor recall for minority class
- Unfair and unreliable predictions

Example

Fraud Detection Dataset

Class Distribution



Model Trained Without Handling Imbalance

		Predicted: Non-Fraud	Predicted: Fraud
Actual	Non-Fraud	99,000	200
	Fraud	750	50

Accuracy : 99.25%
Precision (Fraud) : 20%
Recall (Fraud) : 6.25%
F1-Score (Fraud) : 9.5%

Consequence

Looks accurate but fails on the minority class that matters most.

4 Failing to document feature provenance,

Making later audits and explanations difficult.

Why it's a problem

- No lineage of how features were created
- Hard to reproduce results
- Audit & compliance risk
- Weakens explanations

Example

Feature	Created From	Transformations	Created By	Date
Debt_to_Income_Ratio	Income, Total Debt	Total Debt / Income	Analyst A	12-Apr-24
Credit_Utilization_Rate	Credit Card Limit, Balance	Balance / Limit	Analyst B	12-Apr-24
Customer_Risk_Index	5 Raw Features	Weighted Sum ($w_1..w_5$)	Analyst C	12-Apr-24
...	...	...	...	...

Consequence

Difficult to audit, explain, or track changes; increases compliance risk.

Good EDA is the foundation of good feature engineering. | Clean, interpretable, and well-documented features lead to reliable, fair and explainable AI.

QUESTION BANK – EXPLAINABLE AI (XAI)

L1 – REMEMBER

Q1. Define Explainable Artificial Intelligence (XAI). (BL: L1)

Q2. What is the primary objective of Exploratory Data Analysis (EDA)? (BL: L1)

Q3. List any four feature transformation techniques used in feature engineering. (BL: L1)

Q4. Differentiate between scaling and normalization. (BL: L1)

Q5. Define Principal Component Analysis (PCA). (BL: L1)

Q6. What is One-Hot Encoding? (BL: L1)

Q7. What is the purpose of t-SNE? (BL: L1)

Q8. List the three major feature selection methods. (BL: L1)

L2 – UNDERSTAND

Q9. Explain why EDA is considered an important preprocessing step before machine learning. (BL: L2)

Q10. Explain how scaling improves machine learning model performance. (BL: L2)

Q11. Describe the working principle of PCA with a suitable example. (BL: L2)

Q12. Explain the difference between PCA and t-SNE. (BL: L2)

Q13. Explain how One-Hot Encoding converts categorical variables into numerical form. (BL: L2)

Q14. Describe the working principle of wrapper-based feature selection. (BL: L2)

Q15. Explain the importance of feature engineering in improving model accuracy. (BL: L2)

Q16. Explain the need for explainability in Artificial Intelligence. (BL: L2)

L3 – APPLY

Q17. Apply Min-Max Normalization to the values: 25, 50, 75, 100. (BL: L3)

Q18. A categorical feature "Vehicle Type" contains Car, Bike, Bus. Apply One-Hot Encoding. (BL: L3)

Q19. A dataset contains four correlated features. Demonstrate how PCA can reduce them into two principal components. (BL: L3)

Q20. Given student marks in Mathematics, Physics, Chemistry and English, identify one meaningful interaction feature that can improve prediction performance. (BL: L3)

Q21. A dataset contains two highly correlated features: Height and Arm Length. Recommend an appropriate feature selection method and justify your choice. (BL: L3)

Q22. A computer vision dataset contains 60% dark images and 40% bright images. Suggest suitable EDA techniques to analyze the dataset. (BL: L3)

L4 – ANALYZE

Q23. Analyze the effect of skipping EDA before model development. Discuss its impact on explainability. (BL: L4)

Q24. Analyze the advantages and limitations of PCA and t-SNE for dimensionality reduction. (BL: L4)

Q25. A machine learning model achieves 98% accuracy but provides no explanation for its predictions. Analyze why this may still be unsuitable for healthcare applications. (BL: L4)

Q26. Analyze the impact of poor feature engineering on the interpretability of AI models. (BL: L4)

Q27. A dataset contains class imbalance, missing values and redundant features. Analyze the sequence of preprocessing steps required before model training. (BL: L4)

Q28. Compare Ordinal Encoding, One-Hot Encoding and Target Encoding with respect to their advantages, disadvantages and suitable applications. (BL: L4)

Q29. Analyze how feature engineering contributes to the development of trustworthy and explainable AI systems. (BL: L4)